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RESEARCH ARTICLE

A Novel Multi-Image Encryption Scheme Using Generalized Rectangular Transform and Advanced 5-D Hyperchaotic Map

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ABSTRACT This paper presents GRTPHM (Generalized Rectangular Transform and Penta-Hyperchaotic Map), a novel multi-image encryption scheme that enhances security, efficiency, and versatility. The encryption process consists of two key phases: pixel permutation and pixel diffusion. The pixel permutation phase leverages the Generalized Rectangular Transform (GRT) to rearrange pixel positions using a flexible transformation matrix, effectively disrupting strong correlations between neighboring pixels and improving resistance to statistical analysis. In the pixel diffusion phase, a bit-level random permutation (BLRP) technique is employed alongside a Penta-Hyperchaotic Map (PHM), an advanced 5-D discrete memristor and meminductor-based hyperchaotic map, integrated with the SHA-512 hashing function. Unlike traditional pixel-based diffusion methods, BLRP shuffles bits, achieving finer confusion and making structural inference significantly more difficult for attackers. The PHM generates a highly unpredictable chaotic sequence, ensuring substantial changes in ciphertext even for minor modifications in plaintext or encryption parameters. This enhances the avalanche effect, strengthening resistance to differential and statistical attacks. GRTPHM introduces a novel reverse rectangular transform (IGRT) that overcomes the limitations of existing approaches, enabling efficient and accurate image reconstruction. Notably, decryption does not require prior knowledge of the transformation matrix's periodicity and achieves a speedup of up to 299 times compared to state-of-the-art methods. Extensive robustness testing against cropping, salt & pepper noise, and hybrid attacks confirms that GRTPHM can maintain satisfactory decryption accuracy and visual quality, even under severe image distortions. The results indicate that cropping attacks cause greater degradation than noise-based attacks due to the irreversible loss of contiguous pixel regions. The scheme supports a wide range of image sizes and formats, including rectangular and equilateral images, as well as single and multi-image encryption scenarios. Performance evaluations demonstrate that GRTPHM outperforms existing encryption techniques in terms of accuracy, computational efficiency, security, and adaptability. Comparative analyses confirm its robust encryption and decryption capabilities, effectively protecting against chosen-plaintext and differential attacks. These results establish GRTPHM as a comprehensive and reliable solution for modern image encryption applications.

INDEX TERMS Multi-image encryption, generalized rectangular transform (GRT), bit-level random permutation, a penta-hyperchaotic map (PHM), inverse rectangular transform, chaotic cryptography.

I. INTRODUCTION

In the digital era, ensuring the secure transmission of multimedia content, particularly images, has become crucial due to the increasing reliance on digital communication

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and the susceptibility of such data to unauthorized access and manipulation. Traditional image encryption methods, adapted from text-based encryption algorithms, often fail to account for the unique properties of image data, such as pixel correlation and high data redundancy [1]. Consequently, these methods may not achieve satisfactory performance in terms of security and efficiency.

Recent research has shifted toward developing specialized encryption schemes tailored specifically for images, incorporating concepts from chaos theory and advanced mathematical transformations [2], [3], [4]. Chaotic systems, characterized by their sensitivity to initial conditions and pseudo-random behavior, have proven particularly effective for cryptographic applications. Multi-dimensional chaotic maps provide a higher level of complexity and security due to their expanded key space and intricate dynamical behaviors [5], [6], [7].

An emerging focus in image encryption is the challenge of rectangular image formats [8], [9], [10]. In many real-world scenarios, images do not conform to square or equilateral dimensions, such as panoramic views, scanned documents, or specialized media formats. Existing encryption schemes, typically optimized for square images, may struggle with rectangular formats due to uneven data distribution and varying aspect ratios. This can lead to suboptimal encryption performance and potential vulnerabilities when these methods are scaled or adapted for non-square image sizes. Therefore, it is essential to design encryption schemes that maintain robustness regardless of the aspect ratio or size of the image.

To address these challenges, this paper introduces a novel rectangular multi-image encryption scheme named GRTPHM (Generalized Rectangular Transform and Penta-Hyperchaotic Map). The proposed scheme integrates two main phases that enable effective, secure, and flexible image encryption. In the pixel permutation phase, the Generalized Rectangular Transform (GRT) is used to rearrange pixel positions, breaking the strong correlations between neighboring pixels and enhancing resistance to statistical analysis. The GRT employs a flexible transformation matrix that can be adapted to support both equilateral and rectangular images, significantly enhancing the scheme's applicability to real-world multimedia content.

In the pixel diffusion phase, the scheme utilizes the SHA-512 hashing function, combining it with the plaintext image to produce 512 hash values as part of the encryption key. This process creates a complex, non-obvious relationship between the ciphertext and the encryption key, ensuring that even minor changes in the plaintext (e.g., a one-bit modification) result in substantial changes in the hash, thereby impacting the encrypted image. To modify pixel values, we employ a bit-level random permutation (BLRP) technique, which shuffles bits rather than pixels or bytes, effectively disrupting pixel correlations more than traditional pixel-based methods. This approach introduces a finer level of confusion, significantly complicating an attacker's ability to infer structural information or decipher meaningful patterns. In addition to BLRP, we propose an advanced 5-D discrete memristor and meminductor (PHM) map that generates a hyperchaotic sequence characterized by high randomness and unpredictability. This PHM map disperses the effect of single plaintext pixels, as well as previous encrypted pixels, across the entire ciphertext, amplifying the avalanche effect and greatly enhancing resistance to differential attacks and pattern recognition.

The core objective of this research is to develop a robust and adaptive encryption framework—GRTPHM—that enhances security, computational efficiency, and resistance to modern cryptographic attacks. Unlike conventional chaos-based cryptographic methods that rely on periodicity analysis, the proposed scheme introduces an effective inverse transformation, eliminating the need for periodicity knowledge. The integration of Generalized Rectangular Transform (GRT) ensures strong diffusion characteristics, while the 5-D hyperchaotic system introduces a higher degree of randomness and complexity, making the encryption highly resistant to brute force, differential, and statistical attacks.

Through rigorous theoretical analysis, simulations, and security evaluations, this study aims to demonstrate the superiority of the GRTPHM encryption method over existing schemes in terms of entropy, key sensitivity, correlation coefficient, and computational efficiency. The findings of this research contribute to advancing the field of image security, particularly for applications in secure image transmission, medical imaging, and cloud-based image storage systems.

This paper introduces several significant innovations that enhance image encryption security, efficiency, and adaptability:

- *Generalized Rectangular Transform (GRT) for Flexible Pixel Permutation:* Unlike conventional transforms that work primarily on square images, GRT supports both equilateral and rectangular images, ensuring broad applicability across various multimedia formats. By effectively decorrelating neighboring pixels, GRT strengthens the encryption process against statistical attacks.
- *Penta-Hyperchaotic Map (PHM) for High-Dimensional Security:* The 5-D discrete memristor and meminductor-based hyperchaotic system introduces high-dimensional unpredictability, ensuring strong randomness and robustness against cryptanalytic attacks. The chaotic sequences generated by PHM enhance diffusion characteristics, effectively resisting chosen-plaintext and differential attacks.
- *Bit-Level Random Permutation (BLRP) for Finer Diffusion:* Unlike traditional pixel-based diffusion methods, BLRP operates at the bit level, introducing a higher level of confusion and making statistical inference significantly more difficult. This approach increases encryption security by enhancing sensitivity to minor changes in the plaintext image.

- *Efficient and Adaptive Decryption Process:* The proposed inverse transformation does not require periodicity knowledge, unlike many existing encryption methods, improving decryption accuracy and reliability. The new inverse transformation mechanism achieves a 299× speedup in decryption, significantly outperforming conventional decryption approaches.
- *Multi-Image Encryption Strategy for Enhanced Security and Efficiency:* The proposed scheme is optimized for simultaneous encryption of multiple images, either by reshaping multiple smaller images into a larger rectangular format or combining three grayscale images into a single pseudo-color image. This capability is particularly useful in multimedia applications requiring fast, efficient, and secure multi-image transmission.
- *Superior Performance Compared to Existing Methods:* Security and performance evaluations demonstrate that GRTPHM achieves better entropy, lower correlation coefficients, higher key sensitivity, and enhanced computational efficiency compared to state-of-the-art encryption schemes. Experimental results confirm strong resistance against statistical, differential, and chosen-plaintext attacks, making it a highly secure and efficient encryption solution for real-world applications.

The remainder of this paper is structured as follows: Section II reviews related work, followed by a detailed explanation of the rectangular transformation in Section III. Section IV introduces the proposed GRTPHM encryption scheme, while Section V presents the advanced 5-D discrete memristor and meminductor map. Section VI demonstrates image encryption using GRTPHM with an illustrative example. Section VII evaluates the scheme's security and performance through comprehensive analyses and experimental results. Finally, the paper concludes with key findings and future research directions.

II. RELATED WORKS

Several recent image encryption algorithms employing various techniques, including the Arnold transformation scheme and chaotic sequences, have been introduced in the literatures [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], and [11]. A comprehensive survey of these algorithms can be found in recent studies [8], [9], [10]. To the best of our knowledge, limited research has focused on the encryption of rectangular images. In this section, we review these works to provide deeper insights into our proposed GRTPHM scheme.

While most image encryption algorithms existing in literature focus on processing equilateral images, limited schemes support rectangular image encryption. Zhang et al. [12] proposed a novel chaos-based image encryption scheme utilizing a 2D rectangular transform combined with dependent substitution to scramble rectangular images. Double dependent substitution was implemented to alter pixel values based on pixel positions derived from the rectangular transformation. The scheme comprises two stages of encryption, each integrating confusion and diffusion operations simultaneously through pixel traversal. Consequently, the proposed method achieves high efficiency and satisfactory security performance. Unfortunately, their method does not suggest a general form of transformation matrix.

Wu et al. [13] introduced an innovative encryption approach for rectangular images by combining rectangular transformations with chaotic tent maps. The rectangular transformation rearranges pixel positions, while the chaotic tent maps add a strong layer of unpredictability to the encryption process. This combination is particularly effective because rectangular transformations allow flexible pixel rearrangement, and the chaotic tent maps introduce significant complexity and sensitivity to initial conditions—key features for secure encryption.

Zhu and Sun [14] conducted a detailed analysis of Wu et al.'s algorithm, highlighting potential vulnerabilities that attackers could exploit. They identified three critical weaknesses: susceptibility to known-plaintext attacks (KPA), inability to resist chosen-plaintext attacks (CPA), and failure to avoid transient effects in the chaotic sequence. To address these issues, Zhu and Sun proposed several enhancements, such as adjustments to chaotic maps, modifications to key scheduling, and improvements in the application of the rectangular transformation. Their revised algorithm successfully mitigates these shortcomings, offering a more secure and robust approach to image encryption using rectangular transformations and chaotic maps.

While the algorithms proposed in [12], [13], and [14] employ rectangular transformations to permute pixels, they employ naïve inverse forms to restore pixel positions. This suffers from a critical limitation: the inability to reconstruct the original plaintext image accurately. This is because the inverse matrix must perform modulus operation not only on the x-direction but also on the y-direction. A counter example will be presented in next section. In contrast, our proposed GRTPHM scheme overcomes this drawback, ensuring the success of image decryption. A detailed explanation of GRTPHM is provided in the next section.

III. RECTANGULAR TRANSFORMATION

A. RT AND NAÏVE INVERSE FORM

A common form of the rectangular transform (RT) for encrypting a digital rectangular image of size $M \times N$ us is shown in (1). Here, $R = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ represents the RT matrix, and (r_m, r_n) denotes the shifting vector, which is used to prevent a fixed-point transformation at position $(0, 0)$. The matrix elements (a, b, c, d) must satisfy the condition outlined in (2), where $p = GCD\,(M, N)$, $h = M/p$, and $v = N/p$, where GCD represents the greatest common divisor. A valid RT matrix ensures that periodicity exists, allowing the transformation to map a position (x, y) back to itself.

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} r_m \\ r_n \end{bmatrix} \right\} mod \begin{bmatrix} M \\ N \end{bmatrix} \quad (1)$$

$$\begin{cases} (b \ mod \ h) = 0 \text{ or } (c \bmod v) = 0 \\ GCD\,(G, p) = 1 \\ GCD\,(a, h) = 1 \\ GCD\,(d, v) = 1 \end{cases} \tag{2}$$

A naïve inverse rectangular transform (IRT) is presented in (3), where $R^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1}$ represents the inverse transformation matrix. Theoretically, if the rectangular transform (RT) is applied G times, the original plaintext image can be reconstructed by performing G times of the IRT.

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} \begin{bmatrix} x' - r_m \\ y' - r_n \end{bmatrix} mod \begin{bmatrix} M \\ N \end{bmatrix} \tag{3}$$

Surprisingly, this form of the IRT does not guarantee a successful inverse transformation. This limitation arises because the process involves two modulus operations, and the condition $M \neq N$ (rectangular image) further complicates the reconstruction. To illustrate this issue, we provide an example using the same transformation matrix proposed in [14], demonstrating the unexpected results that are often overlooked in the literature.

Example 1: Let $R = \begin{bmatrix} 1 & 3 \\ 5 & 16 \end{bmatrix}$ be an RT matrix, and consider a rectangular image of size $M \times N = 512 \times 256$. For simplicity, assume the shifting vector $(r_m, r_n) = (0, 0)$. Using one iteration of the RT, the pixel position $(x, y) = (59, 68)$ is transformed to $(x', y') = (263103)$. However, when applying the naïve IRT with the inverse matrix $R^{-1} = \begin{bmatrix} 16 & -3 \\ -5 & 1 \end{bmatrix}$, the pixel position after one iteration is $(x, y)_r = (315, 68)$, which differs from the original pixel position.

Table 1 presents seven examples where the original pixel positions are reconstructed using a naïve inverse matrix based on (3). This table demonstrates that the naïve inverse matrix R^{-1} fails to accurately reconstruct the original pixel positions (as indicated by the red-highlighted pixels). Notable, even when the determinants of the matrices are either 1 or −1, the reconstruction remains unsuccessful.

TABLE 1. RT and faulty inversed results using naïve (NIRT) matrix $M \times N = 768 \times 512$, $(r_m, r_n) = (0, 0)$.

Original (x, y)	Matrix R	Mapping (x', y')	Inv. Mat R^{-1}	NIRT $(x, y)_r$
(192, 241)	$\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$	(433, 83)	$\begin{bmatrix} 3 & -1 \\ -2 & 1 \end{bmatrix}$	(448, 241)
(356, 511)	$\begin{bmatrix} 5 & 13 \\ 8 & 21 \end{bmatrix}$	(20, 339)	$\begin{bmatrix} 21 & -13 \\ -8 & 5 \end{bmatrix}$	(621, 511)
(229, 195)	$\begin{bmatrix} 1 & 5 \\ 2 & 9 \end{bmatrix}$	(436, 165)	$\begin{bmatrix} -9 & 5 \\ 2 & -1 \end{bmatrix}$	(741, 195)
(318, 317)	$\begin{bmatrix} 2 & 27 \\ 1 & 13 \end{bmatrix}$	(747, 343)	$\begin{bmatrix} -13 & 27 \\ 1 & -2 \end{bmatrix}$	(318, 61)
(192, 150)	$\begin{bmatrix} 4 & 21 \\ 1 & 5 \end{bmatrix}$	(78, 430)	$\begin{bmatrix} -5 & 21 \\ 1 & -4 \end{bmatrix}$	(192, 406)
(645, 12)	$\begin{bmatrix} 2 & 3 \\ 27 & 41 \end{bmatrix}$	(558, 499)	$\begin{bmatrix} 41 & -3 \\ -27 & 2 \end{bmatrix}$	(645, 268)
(767, 511)	$\begin{bmatrix} 5 & 13 \\ 38 & 99 \end{bmatrix}$	(494, 375)	$\begin{bmatrix} 99 & -13 \\ -38 & 5 \end{bmatrix}$	(255, 511)

Figure 1 demonstrates three images where a naive inverse RT cannot successfully decrypt the ciphered image. This is because the naive inverse matrix is no longer a bijection function under the modulo operations with different divisor.

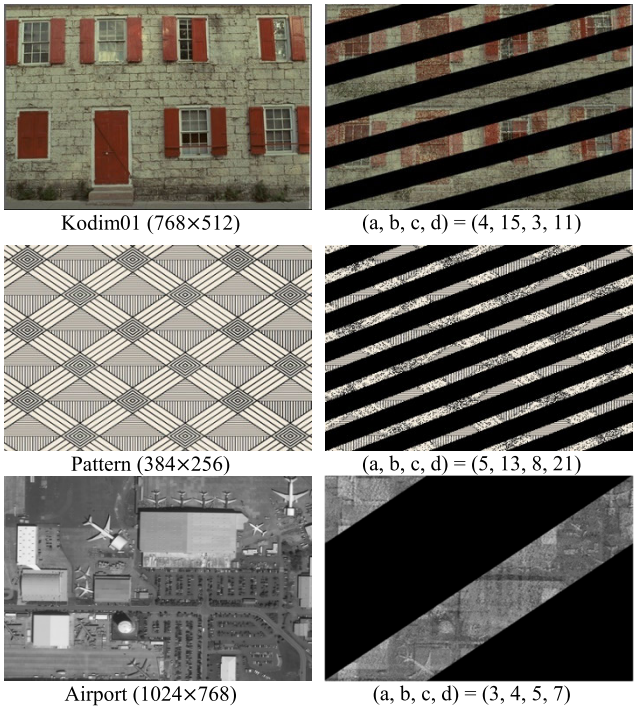


FIGURE 1. The naïve inverse rectangular transformation is unable to accurately reconstruct the original plaintext images.

B. SEPECIAL RT AND ITS INVERSE FORM

In this section, we introduce a special rectangular transformation (SRT) designed to effectively decrypt an image using a corresponding inverse special rectangular transformation (ISRT). The SRT is inspired by a variant of the Arnold transformation matrix. The matrix used in the SRT, denoted by $R = \left[(1, s)^T, (r, rs + 1)^T\right]$, has two parameters, (r, s), with the upper-left element set to a constant 1, as shown in (4). Although the determinant of the matrix is always a unit, two integers r, s must satisfy two constraints, as outlined in (5), where $p = GCD\,(M, N)$, $h = M/p$, $v = N/p$.

$$\begin{bmatrix} x \\ y \end{bmatrix}_{n+1} = \begin{bmatrix} 1 & r \\ s & rs + 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}_n mod \begin{bmatrix} M \\ N \end{bmatrix} \tag{4}$$

$$\begin{cases} (r \ mod \ h) = 0 \text{ or } (s \bmod v) = 0 \\ GCD\,(rs + 1, v) = 1 \end{cases} \tag{5}$$

The pixel reconstruction remains unsuccessful when using a naïve inverse matrix, $R^{-1} = \left[(rs + 1, -s)^T, (-r, 1)^T\right]$. However, accurate reconstruction can be achieved using a special inverse matrix for SRT, as shown in (6). Notably, the reconstruction requires first determining the y-coordinate (y_n), and then using it as an input

to retrieve the x-coordinate (x_n).

$$\begin{cases} y_n = (y_{n+1} - s \cdot x_{n+1}) \, mod \, N \\ x_n = (x_{n+1} - r \cdot y_n) \, mod \, M \end{cases} \tag{6}$$

Table 2 presents a comparison between RT and SRT along with their respective inverse transformation results. The mapping result (x', y') represents the transformed coordinates of a pixel (x, y) after a single RT transformation. Clearly, the naïve inverse matrix fails to restore the original pixel (as indicated by the red-highlighted pixel), despite utilizing an inverse matrix with a unit determinant. In contrast, the inverse SRT successfully reconstructs the original pixel successfully (as indicated by the green-highlighted pixel) using Eq. (6).

TABLE 2. A Comparison of RT and SRT and their inverse results $M \times N = 768 \times 512.(r_m, r_n) = (0, 0)$.

Original (x, y)	Matrix RT	Mapping (x', y')	Naïve Inv. R^{-1}	Restored $(x, y)_r$
(192, 241)	$\begin{bmatrix}1 & 1\\ 2 & 3\end{bmatrix}$	(433, 83)	$\begin{bmatrix}3 & -1\\ -2 & 1\end{bmatrix}$	(448, 241)
(229, 503)	$\begin{bmatrix}1 & 7\\ 8 & 57\end{bmatrix}$	(678, 295)	$\begin{bmatrix}13 & -3\\ -4 & 1\end{bmatrix}$	(485, 503)
(645,506)	$\begin{bmatrix}1 & 11\\ 6 & 67\end{bmatrix}$	(67, 396)	$\begin{bmatrix}57 & -7\\ -8 & 1\end{bmatrix}$	(133, 506)
Original (x, y)	Matrix SRT	Mapping (x', y')	Specific Inv. $ISRT$	Restored $(x, y)_r$
(192, 241)	$\begin{bmatrix}1 & 1\\ 2 & 3\end{bmatrix}$	(433, 83)	(r, s)=(1, 2) Eq. (6)	(192, 241)
(229, 503)	$\begin{bmatrix}1 & 7\\ 8 & 57\end{bmatrix}$	(678, 295)	(r, s)=(7, 8) Eq. (6)	(229, 503)
(645,506)	$\begin{bmatrix}1 & 11\\ 6 & 67\end{bmatrix}$	(67, 396)	(r, s)=(11, 6) Eq. (6)	(645, 506)

We introduce a third integer parameter, $k \geq 0$, alongside (r, s) to expand the range of the transformation matrix, R_E, as shown in (7). It is important to note that the matrix elements must satisfy two constraints, as outlined in (8). The corresponding inverse matrix transformation is shown in (9), where it is evident that when $k = 0$, (9) reduces to (6). Table 3 presents several RT matrices and their corresponding expansions. The matrix expansion offers a key advantage in generating a diverse set of eligible matrices, which can be effectively utilized to shuffle pixel positions in the pixel permutation process.

$$\begin{bmatrix} x \\ y \end{bmatrix}_{n+1} = \begin{bmatrix} 1 & r \\ s+k & r(s+k)+1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}_n mod \begin{bmatrix} M \\ N \end{bmatrix}, \tag{7}$$

$$\begin{cases} r \, mod \, h = 0 \text{ or } (s+k) \mod v = 0 \\ GCD\,(r(s+k)+1, v) = 1 \end{cases} \tag{8}$$

$$\begin{cases} y_n = (y_{n+1} - (s+k) \cdot x_{n+1}) \; mod \, N \\ x_n = (x_{n+1} - r \cdot y_n) \, mod \, M \end{cases} \tag{9}$$

Table 4 shows a comparison between RT, expanded SRT (ESRT), and their inverse results (IESRT). As expected, the naïve inverse matrix fails to reconstruct the original pixel position, as the x-coordinate remains incorrect, despite using the inverse matrix with a unit determinant. In contrast, the IESRT successfully restores the original pixel with high accuracy.

TABLE 3. Some matrices for rectangular transformation and its extension.

No	M × N	$(1, r, s, rs+1)$	k	R_E
1	260×176	(1, 1, 44, 45)	44	(1, 1, 88, 89)
2	260×176	(1, 3, 44, 133)	44	(1, 3, 88, 265)
3	512×384	(1, 1, 30, 31)	33	(1, 1, 63, 64)
4	768×512	(1, 5, 2, 11)	14	(1, 5, 16, 81)
5	768×512	(1, 7, 8, 57)	4	(1, 7, 12, 85)
6	768×512	(1, 7, 17, 120)	15	(1, 7, 32, 225)
7	768×512	(1, 9, 8, 73)	22	(1, 9, 30, 271)

TABLE 4. A Comparison of RT and expanded SRT (ESRT) and their inverse results, $M \times N = 768 \times 512$. $(r_m, r_n) = (0, 0)$.

Original (x, y)	Matrix RT	Mapping (x', y')	Naïve Inv. R^{-1}	Restored $(x, y)_r$
(192, 241)	$\begin{bmatrix}1 & 5\\ 16 & 81\end{bmatrix}$	(629, 65)	$\begin{bmatrix}81 & -5\\ -16 & 1\end{bmatrix}$	(704, 241)
(229, 503)	$\begin{bmatrix}1 & 7\\ 12 & 85\end{bmatrix}$	(678, 447)	$\begin{bmatrix}85 & -7\\ -12 & 1\end{bmatrix}$	(741, 503)
(645,506)	$\begin{bmatrix}1 & 7\\ 32 & 225\end{bmatrix}$	(645,506)	$\begin{bmatrix}225 & -7\\ -32 & 1\end{bmatrix}$	(389, 506)
Original (x, y)	Matrix $ESRT$	Mapping (x', y')	Expand Inv. $IESRT$	Restored $(x, y)_r$
(192, 241)	$\begin{bmatrix}1 & 5\\ 16 & 81\end{bmatrix}$	(629, 65)	(r, s, k) = (5, 2, 14), Eq. (9)	(192, 241)
(229, 503)	$\begin{bmatrix}1 & 7\\ 12 & 85\end{bmatrix}$	(678, 447)	(r, s, k) = (7, 8, 4) Eq. (9)	(229, 503)
(645,506)	$\begin{bmatrix}1 & 7\\ 32 & 225\end{bmatrix}$	(645,506)	(r, s, k) = (7, 17, 15) Eq. (9)	(645, 506)

Table 2 and Table 4 indicates that although SRT and ESRT can faithfully reconstruct the original pixel positions using explicit inverse forms, their transformation matrices are constrained due to the first element being fixed at one, resulting in a maximum of only two degrees of freedom. leading. This limited flexibility makes these schemes more susceptible to malicious attacks.

To address these limitations, the next section introduces the Generalized Rectangular Transform (GRT), which overcomes these constraints and enhances security.

IV. GENERALIZED RECTANGULAR TRANSFORMATION

A. GENERAL FORM

A generalized rectangular transformation (GRT) involves four parameters in the matrix of the form, $G = \left[(a, c)^T, (b, d)^T\right]$ [15]. The GRT, supporting four degrees of freedom, is described in (10), where $(a, b, c, d) \in Z_M^+$. Similarly, the four parameters must satisfy four conditions, as listed in (2), to ensure the existence of periodicity for decryption. It is important to note that the determinant, $\|G\|$, can be either positive or negative, and it is necessarily equal to 1. For example, the matrix, $G = \left[(5, 10)^T, (7, 21)^T\right]$ (with $G = 35$) can be used to encrypt an image of size 768×512. Additionally, an offset $(r_m, r_n) \in Z_M^+$ can be introduced to

prevent the fixed-point transformation effect, as the pixel position [0, 0] would otherwise always map to itself without any change.

We remark that although the authors in [13] and [14] described the four conditions of the matrix, they mistyped two expressions in (4) and (3). In particular, $p_m = p/M$ and $p_n = p/N$ should be rectified to $p_m = M/p$ and $p_n = N/p$, where M and N represent the height and the width of the plaintext image, respectively.

$$\begin{bmatrix} x \\ y \end{bmatrix}_{n+1} = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}_n + \begin{bmatrix} r_m \\ r_n \end{bmatrix} \right\} mod \begin{bmatrix} M \\ N \end{bmatrix} \tag{10}$$

We can extend the coverage of the transformation matrix in the generalized rectangular transformation (GRT). Let $G = \left[(a, c)^T, (b, d)^T\right]$ be the original GRT matrix. By utilizing the properties of the greatest common divisor (GCD), we extend $\boldsymbol{G}$ by multiplying two integers, (e, f), respectively, to create two types of the extended matrix, E_1 and E_2, as shown in (11). In this extension, $e, f \geq 1$ with $GCD(e, M) = 1$ and $GCD(f, N) = 1$. Recall that since $GCD(M, N) = p, h = \frac{M}{p}$, and $v = \frac{N}{p}$, we know that $(e, h), (e, p), (f, v)$, and (f, p) are coprime.

$$E_1 = \begin{bmatrix} ea & eb \\ fc & fd \end{bmatrix} \text{ or } E_2 = \begin{bmatrix} ea & fb \\ ec & fd \end{bmatrix} \tag{11}$$

By examining the four prerequisites, we can prove that E_1 and E_2 are indeed valid GRT transformation matrices, as detailed below.

Condition 1: $(eb \ mod \ h) = 0 \text{or} \ (fc \bmod v) = 0$

This statement is correct as $e, f \geq 1$.

Condition 2: $GCD(efG, p) = 1$

Since the GCD is a multiplicative function, we can break the expression: $GCD(efG, p) = GCD(G, p) \cdot GCD(ef, p)$ and simplify it to $GCD(ef, p)$ because $GCD(G, p) = 1$. Furthermore, since $(e, p), (f, p)$ are coprime, we can further simplify the expression to $GCD(ef, p) = GCD(e, p) \cdot GCD(ef, p) = 1$.

Condition 3 and 4: Recall that since $(e, h), (f, v)$ are coprime, $GCD(ea, h) = GCD(e, h) \cdot GCD(a, h) = 1$. This leads to $GCD(ea, h) = 1$ and $GCD(fd, v) = 1$.

Table 5 presents several RT matrices and their determinant, $\|G\|$, expanded by the original valid matrix, $(a, b, c, d) = (5, 7, 10, 21)$ using different (e, f) for an image of size $M \times N = 768 \times 512$. The expanded matrices make it more difficult for a malicious attacker to deduce the transformation matrix, thereby providing higher security compared to using the original matrix.

B. GENERAL INVERSE FORM

In this section, we introduce an inverse generalized rectangular transformation (IGRT) to resolve the problem of the naïve IRT scheme. Taking the inverse matrix as granted, the image decryption is not discussed by most image encryption algorithms. However as shown in Example 1 and tables, a naïve inverse form of the matrix cannot restore the pixel correctly.

TABLE 5. Some matrices expanded by the base matrix ($M \times N = 768 \times 512$).

No.	(e, f)	Type	(a, b, c, d)	$\|G\|$
base	(1, 1)	E_1	(5, 7, 10, 21)	35
1	(5, 7)	E_1	(25, 35, 70, 147)	1225
2	(7, 19)	E_2	(35, 133, 70, 399)	4655
3	(11, 11)	E_1	(55, 77, 110, 231)	4235
4	(13, 13)	E_2	(65, 91, 130, 273)	5915
5	(17, 23)	E_1	(85, 119, 230, 483)	13685
6	(19, 15)	E_2	(95, 105, 190, 315)	9975
7	(23, 17)	E_1	(115, 161, 110, 231)	8855
8	(29, 5)	E_2	(145, 35, 290, 105)	5075

Our IGRT scheme consists of four significant features. First, given a valid RT matrix (G), IGRT provides a bijection to reversely map the transformed pixel position (x', y') back to the original pixel position (x, y). Second, it can decrypt an image without any prior information about the periodicity of the matrix G. Third, it is general enough to decrypt a squared image if we impose the condition of $M = N$. In addition, it can decrypt non-square image even though the transformation matrix has the form of lower or upper triangle (one of the elements is zero in the diagonal direction). Due to the inversion manner, the decryption is much effective and more efficient than the conventional periodic-decryption process. As a result, this scheme is a universal inverse transformation which is powerful enough to support both squared and non-square images and versatile types of transformation matrices.

The idea behind the proposed IGRT scheme is to separate the process into tow inverse transformations since the matrix used in a regular transformation is a matrix superimposition from generalized Arnold transform and triangular transform. The first inverse transformation is the inverse mapping of GAT, where a common divisor, $p = GCD(M, N)$, is adopted in x and y directions. The second one is the inverse mapping of triangular transformation, where we need to consider two cases: the upper triangle with the respect factor, $h = M/p$, or the upper triangle with the respect factor $v = N/p$.

The IGRT scheme consists of four stages including the inverse equilateral mapping, inverse projection mapping, inverse triangular mapping, inverse translation, as detailed below:

Stage 1: Inverse equilateral mapping. We use p as a common divisor to operate the inverse GAT mapping and derive an inverse position $\left[x_p, y_p\right]^T$. As shown in (12), the input is a ciphered pixel position $\left[x', y'\right]^T$, where $G^{-1}_{[p]} = (ad - bc)^{-1}_{[p]}$ denotes the inverse of the determinant modulo p.

$$\begin{bmatrix} x_p \\ y_p \end{bmatrix} = G^{-1}_{[p]} \begin{bmatrix} d & (p-1)\, b \\ (p-1)\, c & a \end{bmatrix} \begin{bmatrix} x' - r_m \\ y' - r_n \end{bmatrix} mod \ p \tag{12}$$

Stage 2: Inverse projection mapping. In this stage, we determine $[H, V]^T$ from $[x', y']^T$ and $[x_p, y_p]^T$, as shown in (13). Note that we need to input $\left[x', y'\right]^T$ and $[x_p, y_p]^T$, which is

just derived from Stage 1.

$$\begin{bmatrix} H \\ V \end{bmatrix} = \begin{bmatrix} x' - r_m \\ y' - r_n \end{bmatrix} - \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_p \\ y_p \end{bmatrix} + \begin{bmatrix} \left\lceil \frac{a \cdot p}{h} \right\rceil * h + \left\lceil \frac{b \cdot p}{h} \right\rceil * h \\ \left\lceil \frac{c \cdot p}{v} \right\rceil * v + \left\lceil \frac{d \cdot p}{v} \right\rceil * v \end{bmatrix} \bmod \begin{bmatrix} h \\ v \end{bmatrix} \quad (13)$$

Stage 3: Inverse triangular mapping. Depending on the results of (b mod h) and (c mod v), we operate the inverse triangular mapping to produce $[x_h, y_v]^T$. There are three distinct cases, as defined in (14), (15), (16), respectively. It is important to note that these three cases can be verified by examining the elements in the transformation matrix.

Case 1: The first case indicates that we only need to consider the mapping in the diagonal direction because the transformation matrix is $\begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix}$, thereby b and c both satisfying the determination condition. Element appearing in the diagonal direction mean that x_h and y_v can be derived independently.

$$\text{Case 1: } b \mod h = 0 \text{ and } c \mod v = 0$$

$$\begin{bmatrix} x_h \\ y_v \end{bmatrix} = \begin{bmatrix} a_{[h]}^{-1} \cdot H \\ d_{[h]}^{-1} \cdot V \end{bmatrix} mod \begin{bmatrix} h \\ v \end{bmatrix} \quad (14)$$

The second case indicates that we consider a lower triangle because the matrix is $\begin{bmatrix} a & 0 \\ c & d \end{bmatrix}$, which means that we must obtain x_h because y_v depends on it.

$$\text{Case 2: } b \mod h = 0 \text{ and } c \mod v \neq 0$$

$$\begin{bmatrix} x_h \\ y_v \end{bmatrix} = \begin{bmatrix} a_{[h]}^{-1} \cdot H \\ d_{[v]}^{-1}(V + \left\lceil \frac{c \cdot h}{v} \right\rceil v - c \cdot x_h) \end{bmatrix} + mod \begin{bmatrix} h \\ v \end{bmatrix} \quad (15)$$

The third case means that we consider an upper triangle because the matrix is $\begin{bmatrix} a & b \\ 0 & d \end{bmatrix}$. In this case, we need to derive y_v first as x_n depends on y_v.

$$\text{Case 3: } b \mod h \neq 0 \text{ and } c \mod v = 0$$

$$\begin{bmatrix} y_v \\ x_h \end{bmatrix} = \begin{bmatrix} d_{[v]}^{-1} \cdot V \\ a_{[h]}^{-1}(H + \left\lceil \frac{b \cdot v}{h} \right\rceil h - b \cdot y_v) \end{bmatrix} mod \begin{bmatrix} v \\ h \end{bmatrix} \quad (16)$$

Stage 4: Inverse translation. We add the respective translation vectors, $p \cdot x_h$ and $p \cdot y_v$, to the inverse position derived in the first stage, thus producing the completely inverse position, $[H, V]^T$, as shown in (17).

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x_p \\ y_p \end{bmatrix} + \begin{bmatrix} p \cdot x_h \\ p \cdot y_v \end{bmatrix} \quad (17)$$

We now adopt parameter shown in Example 1 to show that our IGRT scheme can correctly product the inversed position.

Example 2: Let $\boldsymbol{G} = \left((1,5)^T, (3,16)^T\right), M \times N = 512 \times 256$ and $(r_m, r_n) = (0,0)$. A pixel position $(x, y) = (59, 68)$ is mapped to $\left(x', y'\right) = (263, 103)$ through the RT transformation. We now adopt the IGRT to produce the inversed position. To begin with, we first derive three intermediate parameters: $p = 256, h = 2$, and $v = 1$. In addition, we can derive the following values which can be used later include $G_{[p]}^{-1} = 1, a_{[p]}^{-1} = 1$, and $d_{[v]}^{-1} = 0$.

Stage 1: Given $(a, b, c, d) = (1, 3, 5, 16)$, we first derive the determinant $G = 1$ and $G_{[p]}^{-1} = 1$. Next, referring to (12), we obtain $\left(x_p, y_p\right) = (59, 68)$.

Stage 2: We use the derived value in the previous state as an input and obtain (H, V) = (0, 0), according to (13).

Stage 3: Since b mod $h \neq 0$ and c mod $v = 0$, we adopt the expression in case 3. First, we calculate $y_v = 0$ and then we can input y_v in (16) to derive $x_h = 0$.

Stage 4: Referring to (17), we obtain the final inverse position (x, y) = (59, 68), which is identical to the original input position.

Table 6 shows the correct results derived by our IGRT scheme using the identical parameters including the same transformation matrices, R, shown in Table 1. We list the intermediate results at each stage. Observing the statistics, our IGRT scheme can produce correct inverse pixel position through four-stage process.

TABLE 6. GRT and correct IGRT results, $M \times N = 768 \times 512$.

Original (x, y)	Stage 1 (x_p, y_p)	Stage 2 (H, V)	Stage 3 (x_h, y_v)	Stage 3 Case	Stage 4 (x, y)
(192, 241)	(192, 241)	(0, 0)	(0, 0)	3	(192, 241)
(365, 511)	(109,255)	(0, 1)	(1, 1)	3	(365, 511)
(229, 195)	(229, 195)	(0, 0)	(0, 0)	3	(229, 195)
(318, 317)	(62, 61)	(2, 0)	(1, 1)	2	(318, 317)
(192, 150)	(192, 150)	(0, 0)	(0, 0)	2	(192, 150)
(645, 12)	(133, 12)	(1, 0)	(2, 0)	2	(645, 12)
(767, 511)	(253, 255)	(2, 1)	(2, 1)	3	(767, 511)

V. PHM: ADVANCED 5-D DISCRETE MEMRISTOR AND MEMINDUCTOR

Pixel diffusion changes values after the pixel is permuted to its new position. In this paper, we make use of hyperchaotic map to alter the pixel values. Recently, Liu et al. [5] presented an approach to producing a new hyperchaotic map based on discrete memristor and meminductor. They are fundamental circuit components that exhibit unique, non-linear properties. A memristor is a two-terminal passive electronic component whose resistance varies based on the history of the electrical charge that has passed through it. This property allows it to "remember" past current levels even when the power is turned off. In recent years, researchers have increasingly incorporated memristors into traditional artificial neural networks, achieving significant breakthroughs in areas such as image recognition, emotional learning, and associative memory [35]. Beyond image encryption, memristors have also been employed for video encryption [30], [31].

A meminductor, on the other hand, is an inductor with memory, meaning that its inductance depends on the history of the current that has flowed through it. Figure 2 illustrates the structure of 5-D memristive mapping [5].

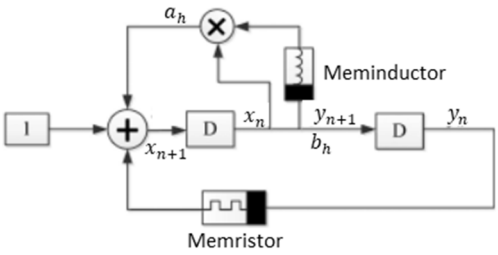


FIGURE 2. Structure diagram of chaotic mapping utilizing discrete memristors and discrete meminductors.

Liu et al. applied both the discrete memristor and discrete meminductor to Henon mapping, thereby obtaining a 5-D chaotic mapping, as shown in (18). An image encryption algorithm was proposed making use of 2^{18} chaotic numbers to encrypt three 256×256 images. Their simulation results show that the image encryption scheme based on the hyperchaotic mapping has better security performance.

$$\begin{aligned} x_{n+1} &= 1 + [c_h + d_h * sin\,(u_n)]\, y_n \\ &\quad - [a_h e_h + a_h g_h * \cos(w_n)] x_n z_n \\ y_{n+1} &= b_h * x_n \\ z_{n+1} &= z_n + k * x_n \\ w_{n+1} &= w_n + k * z_n \\ u_{n+1} &= u_n + k * y_n \end{aligned} \tag{18}$$

Unfortunately, when we utilized the parameters suggested in [5] to generate a longer sequence ($> 2^{19}$), we found that the floating-point values in z_n, w_n, u_n are larger than the maximal upper bound. To overcome this drawback, we instead present an advanced 5-D discrete memristor and meminductor (PHM), as shown in (19). It represents a 5-D hyperchaotic mapping with the division calculation, still taking advantage of five variables and maintaining the features of hyperchaotic sequence.

$$s_{n+1} = [(z_{n+1} * w_{n+1})/(x_{n+1} * y_{n+1} * u_{n+1})]_{17:24} \tag{19}$$

When different parameters are set while keeping the initial values fixed at $x_0 = -10, y_0 = 5, z_0 = 10, w_0 = 1, u_0 = -5$, the phase diagram of the PHM map is shown in Fig. 3. Clearly, PHM can generate diverse chaotic attractors, demonstrating greater flexibility compared to conventional chaotic maps.

Measuring Lyapunov exponents (LEs) is essential for analyzing and understanding the dynamics of nonlinear systems, especially in the study of chaotic and hyperchaotic systems. LEs quantify the average exponential rate of divergence or convergence of adjacent trajectories in the system's phase space over time. A system exhibits chaotic behavior when it has at least one positive LE. Furthermore, when a system possesses more than two positive LEs, it demonstrates hyperchaotic behavior, which offers enhanced unpredictability, complexity, and security compared to chaotic systems.

The bifurcation diagram and Lyapunov exponents are shown in Fig. 4 for parameter values $b \in [-2, 2]$ or

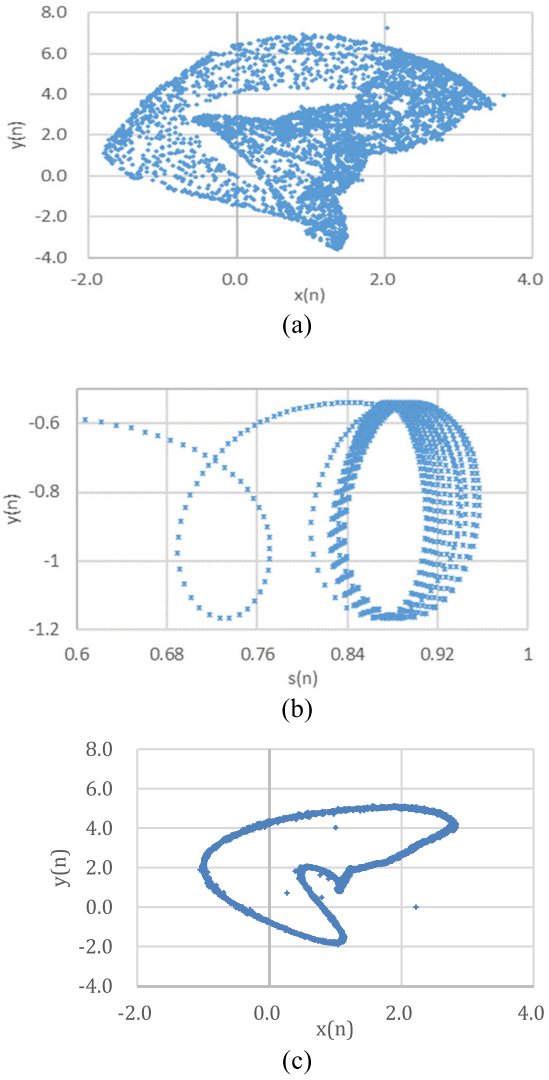


FIGURE 3. Phase diagram of the chaotic map; (a) a_h= 0.0008, b_h= 2.0, c_h= 0.09, d_h= 0.5, e_h= 0.0001, g_h= 0.01, k= 0.3; (b) a_h= 0.0002, b_h=−0.8, c_h= 0.1, d_h= 0.5, e_h= 0.01, g_h =0.01, k= 0.11. $s(n)$ represents the floating-point value before extracting bit information to derive the corresponding integer. (c) all parameters are the same as in (a) except b_h= 1.75.

$b \in [0, 2]$. The remaining parameters are set as follows: $a_h = 0.0008, c_h = 0.09, d_h = 0.5, e_h = 0.0001, g_h = 0.01$, and $k = 0.3$. The initial value is $x_0 = -10, y_0 = 5, z_0 = 10, w_0 = 1, u_0 = -5$. As observed in the figure, the u(n) consistently exhibits a positive Lyapunov exponent across all values of b, confirming chaotic behavior. Additionally, $x(n)$ and $y(n)$ display positive Lyapunov exponent within two specific intervals: a negative range around $b \in [-2, -1.5]$ and a positive interval around $b \in [1, 2]$. In these two intervals, the system exhibits a hyperchaotic state, as evidenced by the presence of three positive Lyapunov exponents. Furthermore, the bifurcation diagrams of $x(n)$ and $s(n)$ as functions of b are also depicted in Fig. 4. It is evident that the phase diagram of chaotic attractor varies with

parameter b. The analysis of Lyapunov exponents and phase diagrams confirms that the discrete chaotic system possesses rich dynamic characteristics.

We test the randomness of our proposed PHM using the National Institute of Standards and Technology (NIST) SP800-22 [16]. We prepare our test set as follows. First, to avoid the transient effect, we utilize the t-th number from hyperchaotic sequence shown in (19), where $t \geq 12000$. Next, we convert each output in $\{s_i\}$ to a 64-bit binary stream using IEEE 754 floating-point standard and retrieve the digital numbers from 17-th to 24-th in a binary stream. Finally, we generate 120 hyperchaotic sequences and ensure that the length of each sequence is greater than 10^6 bits, complying with the prerequisite needed for the NIST testing.

Table 7 lists the results of NIST SP800-22 over 120 sequences by setting the significance level $\alpha = 0.01$, as suggested. A binary sequence can pass a subtest if its generated P-value is larger than or at least equals to the significant level. As can be seen, P-values in most test cases are in line with the requirements. In addition, all subset passes except no. 5, 13, 14, and 15, which have, at most, five failed cases out of 120. The NIST test proves that the proposed PHM sequence has superior randomness performance.

TABLE 7. NIST SP800-22 randomness test results.

No.	Test	P-value	Result
1	Frequency	0.1633	All Pass
2	Block Frequency	0.7714	All Pass
3	Runs	0.3586	All Pass
4	Longest Runs in a Block	0.7049	All Pass
5	Binary Matrix Rank	0.0429	Pass (118)
6	Discrete Fourier Transform	0.4741	All Pass
7	Non-overlap. Template Match.	0.8364	All Pass
8	Overlap. Template Matching	0.3047	All Pass
9	Universal Statistical Test	0.1938	All Pass
10	Linear Complexity	0.4735	All Pass
11	Serial (ave. two states)	0.9457	All Pass
12	Approximate. Entropy	0.7322	All Pass
13	Cumulative Sums Forward	0.0947	Pass (117)
14	Cumulative Sums Backward	0.1902	Pass (115)
15	Rand. Excur. (ave. 8 states)	0.2819	Pass (119)
16	Rand. Excur.Var. (ave. 18 states)	0.3900	All Pass

VI. IMAGE ENCRYPTION/DECRYPTION USING GRTPHM

A. ALGORITHM OUTLINE

Figure 5 illustrates the encryption process, which comprises four main procedures. Our scheme supports multi-image encryption by allowing the superimposition of three grayscale images into a superimposed pseudo image (I_p) for encryption. The process begins with the adoption of the SHA-512 hash function to generate a 512-bit digest, which is subsequently utilized in the following three procedures.

In the pixel permutation stage, secret keys (detailed in detail in a later section) are used as inputs, and the

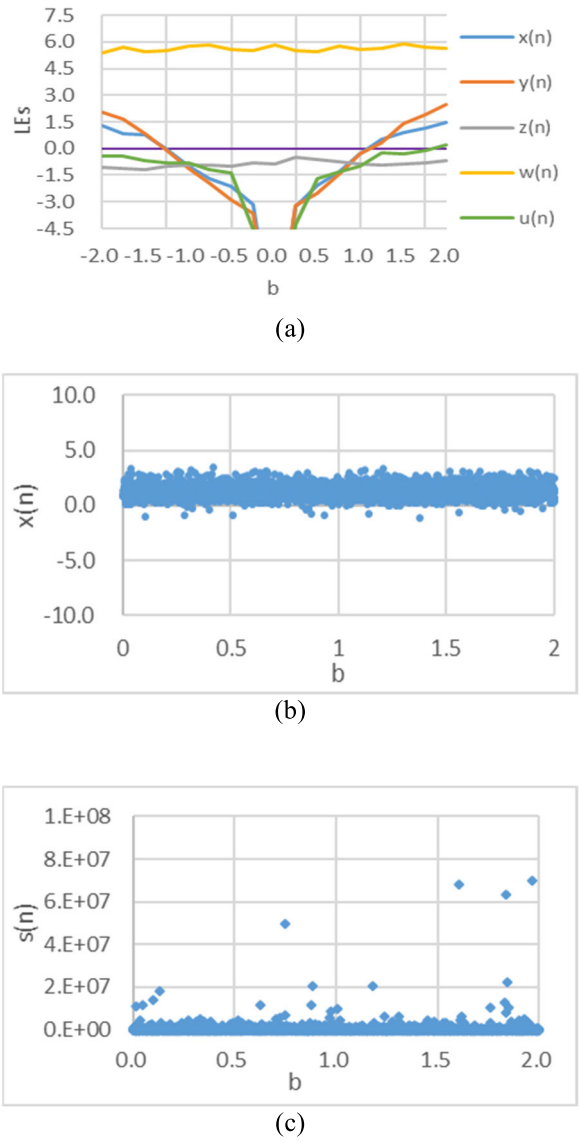


FIGURE 4. Lyapunov exponents and bifurcation diagrams with parameter b. (a) Lyapunov exponents spectrum. (b) Bifurcation diagram for x (n). (c) Bifurcation diagram for s (n).

Generalized Rectangular Transform (GRT) is applied to rearrange pixel positions. Following this, in the pixel diffusion stage, a Bit-Level Random Permutation (BLRP) is employed to shuffle bit positions within each pixel in a pseudo-image containing 24-bit information. Finally, the proposed advanced 5-D discrete memristor and meminductor (PHM) system is utilized to encrypt the pixels, generating the ciphertext image I'_E.

Figure 6 illustrates the image decryption process, which involves three key procedures. First, the PHM system is applied with the decryption formula to decrypt the ciphertext pixels. Next, an inverse bit-level random permutation (IBLRP) is performed to reconstruct each pixel's original value. Finally, the inverse generalized rectangular transform (IGRT) is utilized to restore the pixel positions, resulting in

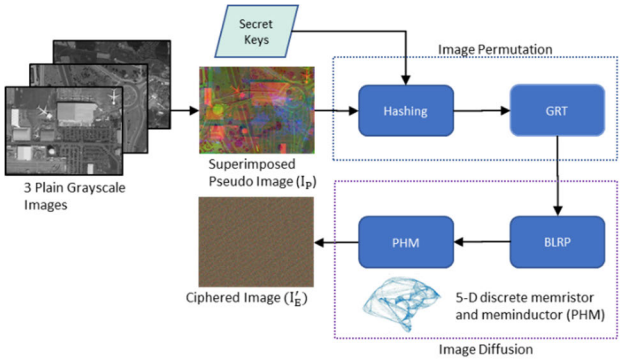


FIGURE 5. Flowchart of the image encryption process, consisting of image permutation and image diffusion.

a superimposed pseudo-plaintext image. From this pseudo image, three grayscale plaintext images can be extracted, thereby completing the decryption process.

B. IMAGE ENCRYPTION

This section outlines the steps of our proposed image encryption/decryption scheme. We assume three plaintext grayscale images have been superimposed, forming a superimposed pseudo image (I_p) with a resolution of $M \times N$. In our scheme, a hash function, hashlib.sha512(I_p), is utilized to generate a 512-bit digest, represented as digest $= b_1 b_2 \ldots b_{512}$. Subsequently, a portion of these bits, as shown in (20), is extracted as secret keys, ensuring that the key is dependent on the three plaintext images. For example, we extract 8 bits (17th-24th) from the digest and convert them into the corresponding integer, denoted by $hv_{[17:24]}$.

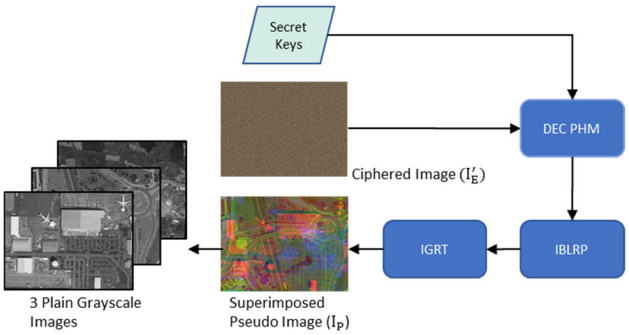


FIGURE 6. Flowchart of the image decryption process, consisting of three procedures.

This approach provides a significant security advantage, as hashing ensures that even a single-bit difference in the input pseudo-image produces a completely distinct hash value. As a result, malicious attackers cannot deduce the encryption rules or infer the key by analyzing the encryption results of multiple original images. Furthermore, this prevents attackers from breaking the encryption algorithm through observation or statistical analysis.

The pixel encryption process in our GRTPHM algorithm contains five steps, as detailed below:

Step 1: Hashing

We input the pseudo-image I_p to the hash function, generating a digest $= b_1 b_2 \ldots b_{512}$.

Step 2: Generalized Rectangular Transform (GRT)

We first select the secret keys, $\{a, b, c, d, e, f, K, r_m, r_n\}$, for the rectangular transformation, where (a, b, c, d) define the transformation matrix, (e, f) are used to expand matrix, K represents the number of GRT iterations, and r_m, r_n are pixel shifting values to present singularity at the coordinate $(0, 0)$. Next, we shuffle the pixel positions by applying generalized rectangular transformation (GRT) for K iterations, producing the GRT image, I_G.

Step 3: Bit-Level Random Permutation (BLRP)

We shuffle the bit positions within each 24-bit pixel in I_p using a random permutation algorithm [38]. The corresponding pseudo-code is provided in Table 8. In this study, we use a fixed seed to generate the random indices j for bit shuffling. This approach ensures that the exact sequence of indices used for swaps can be deterministically reproduced. By using the same seed during decryption, we can reconstruct the shuffled array and reverse the sequence of indices j to restore the original array.

TABLE 8. Pseudo code for the random permutation algorithm.

```
function RanPer(array, r)
  n = length(array)
  for i from n-1 down to 1
    j = random seed (r)   # j ∈ [0, i]
    if   i ≠ j, swap array[i] and array[j]
    else next i
  return array
```

To illustrate the bit-shuffling process, consider an array with five elements, [a, b, c, d, e], indexed from 0 to 4, starting with the last index $i = 4$.

First, we randomly select $j \in [0, 4]$. Referring to r, let $j = 2$. We swap elements at indices 4 and 2, resulting in [a, b, e, d, c].

Next, we start with $i = 3$. We randomly select $j \in [0, 3]$. Referring to r, let $j = 1$. We swap elements at indices 3 and 1, leading to [a, d, e, b, c].

Third, we start with $i = 2$. We randomly select $j \in [0, 2]$. Referring to r, let $j = 0$. We swap elements at indices 2 and 0, leading to [e, d, a, b, c].

Fourth, we start with $i = 1$ and randomly select $j \in [0, 1]$. Referring to r again, let $j = 1$. Since $i = j$, no swap is performed.

After these four steps, the shuffled array becomes [e, d, a, b, c]. If we consider a five-bit array, [1, 0, 1, 1, 0], we can randomly shuffle each element, resulting in [1, 0,bib:1,bib:1,0].

As another example using an 8-bit pixel representation, let the random seed random seed $r = 18$ and assume $p = (231, 125, 29)$ from the superimposed image, where each value is represented as ab 8-bit integer. We store these

bits in an array of 24 elements and perform BLRT, selecting 23 random indices to generate the bit-level permutation result, $p' = (245, 215, 27)$.

BLRP offers four key benefits. First, unbiased process. Each element is shuffled with has an equal probability, ensuring fairness in the permutation. Next, in-place execution. BLRP operates in an in-place manner, modifying the array in place without requiring additional memory. Third, efficient complexity. The algorithm iterates over the array once, resulting in a time complexity of $O(n)$. Finally, reversible transformation. By storing the random seed and performing the random set in reverse order, we can reconstruct the original sequence using Inverse BLRP (IBLRP).

Following our previous example with the resultant order [e, d, a, b, c], we illustrate IBLRP as follows:

First, we consider the first two elements [e, d] at indices 0, 1 in [e, d, a, b, c]. We retrieve the last random index, $j = 1$. Since the element d is at index 1, no swap is needed.

Next, we consider the first three elements [e, d, a] at indices 0, 1, 2 in [e, d, a, b, c]. We retrieve the random index $j = 2$, so we swap elements at index 2, producing [a, d, e, b, c].

Third, we consider the first four elements [a, d, e, b] at indices 0, 1, 2, 3 in [a, d, e, b, c]. We retrieve the random index $j = 1$, so we swap elements at index 1, producing [a, b, e, d, c].

Fourth, we consider the entire elements [a, b, e, d, c] at indices 0, 1, 2, 3, 4. We retrieve the random index $j = 2$, so we swap elements at index 2, producing [a, b, c, d, e].

By following this process, IBLRP successfully reconstructs the original order, provided that the random seeds are recorded and retrieved in reverse order during decryption.

Step 4: Advanced 5-D Discrete Memristor and Meminductor System (PHM)

We select secret keys, including the initial values, $\{x_0, y_0, z_0, w_0, u_0\}$, seven parameters, $a_h, b_h, c_h, d_h, e_h, g_h$, and k, to generate an advanced PHM hyperchaotic sequence, $\{S_{3i}\}$. To enhance security, we utilize digests produced by the SHA-512 algorithm, which are derived from the three plaintext images, thereby significantly increasing resistance to cryptographic attacks. The parameters used are defined in (20). For example, we extract eight bits (17th to 24th) from the digest and convert them into the corresponding decimal value, $hv_{[17:24]}$, which is set as the default parameter a_h. Note that $\{a_h, b_h, c_h, d_h, e_h, g_h\}$ and $k = 0.11$ are used in (18) and (20).

$$\begin{cases} a_h = 0.0002 + 0.5(hv_{[17:24]}/2^8) \times 10^{-4} \\ b_h = 1.0 + 0.5(hv_{[25:32]}/2^8) \\ c_h = 0.1 + 0.5(hv_{[33:40]}/2^8) \times 10^{-1} \\ d_h = 0.5 + 0.5(hv_{[41:48]}/2^8) \times 10^{-1} \\ e_h = 0.01 + 0.5(hv_{[49:56]}/2^8) \times 10^{-2} \\ g_h = 0.01 + 0.5(hv_{[57:64]}/2^8) \times 10^{-2} \end{cases} \tag{20}$$

When an element, $\{S_{3i}\}$, in the hyperchaotic sequence is produced, we perform the pixel encryption simultaneously. First, we retrieve every digital number (17)-th to 24-th bits represented by IEEE 754 floating-point standard), and convert them to the corresponding decimal value between [0, 255]. This value is then used to encrypt a 24-bit pixel within a superimposed image, where 8-bit each in (R_i, G_i, B_i) to produce a ciphertext pixel, (R'_i, G'_i, B'_i) using (21), where $\oplus$ represents the "XOR, exclusive or" operation. The initial values (R'_0, G'_0, B'_0) are adopted 24 digests (8 bits each, from hv_{65} to hv_{88}.

$$p' = \begin{cases} R'_{i+1} = S_{3i} \oplus [(R_{i+1} + B'_i) \ mod \ 256] \\ G'_{i+1} = S_{3i+1} \oplus [(G_{i+1} + R'_i) \ mod \ 256] \\ B'_{i+1} = S_{3i+2} \oplus [(B_{i+1} + G'_i) \ mod \ 256] \end{cases} \tag{21}$$

We remark that the pixel encryption offers a cross-channel operation to produce a ciphertext pixel. For example, we utilize the plaintext pixel in the red channel, R_{i+1}, the ciphertext pixel in the blue channel, B'_i, and the hyperchaotic sequence, S_{3i}, to produce the cipher pixel, R'_{i+1}. This offers a benefit of chain protection across two channels, thereby resisting the differential attack which normally utilizes a whole-black or a whole-white image to unveil potential regular patterns leaking during the encryption process.

C. AN EXAMPLE FOR IMAGE ENCRYPTION

Figure 7 illustrates an example of the image encryption process. Assume a superimposed image (I_p) with a resolution of 768×512. In the first process, Hashing, the image I_p is input into the hash function, generating a 512-bit digest, which will be used later in the PHM process. Next, GRT process. Given an initial pixel position $[x, y] = [192, 241]$, we apply M, $[r_m, r_n]$, and K to transform its coordinates to $[x, y] = [621, 170]$.

In the third process, BLRP, a bit-level permutation is applied to modify the pixel value from form $p = (231, 125, 29)$ to $p = (245, 215, 27)$. In the final PHM process, three hyperchaotic sequence, $S_{3702}, S_{3703}, S_{3704}$, are generated using part of the digest and a previously encrypted pixel $(42, 68, 139)$. These values are then used to encrypt the current pixel, producing the final ciphered pixel $p' = (0, 12, 178)$. By following this encryption process for all pixels in I_p, the final encrypted image I'_E is generated.

D. IMAGE DECRYPTION

The image decryption is the reverse process of the image encryption. We first discuss the secret keys needed in our scheme before detailing the image decryption algorithm.

Table 9 shows nine secret keys used in our scheme. While some are directly derived from the SHA-512 Hashing output, others are assigned by the cipher side. The total bit length required for our scheme is 14 integers and 12 floating-point values, equivalent to 832 bits.

The image decryption, as illustrated in Fig. 6, begins by utilizing the secret keys (keys 6-10) and the DEC PHM

TABLE 9. Secret keys used in our scheme with the given $M \times N$.

No.	Description	Notation	Type	Bit
1	Trans. Matrix	$\{a, b, c, d\}$	4 integers	128
2	Matrix Exp.	$\{e, f\}$	2 integers	64
3	Pixel Shifting	$\{r_m, r_n\}$	2 integers	64
4	Iteration No.	K	1 integer	32
5	Rand. Seed	r	1 integer	32
6	Starting Pos.	t	1 integer	32
7	Initial Values	$\{x_0, y_0, z_0, w_0, u_0\}$	5 flt. Pt.	160
8	Chao. Param.	$\{a_h, b_h, c_h, d_h, e_h, g_h, k\}$	7 flt. Pt.	224
9	Initial Pixels	(R'_0, G'_0, B'_0)	3 integers	96

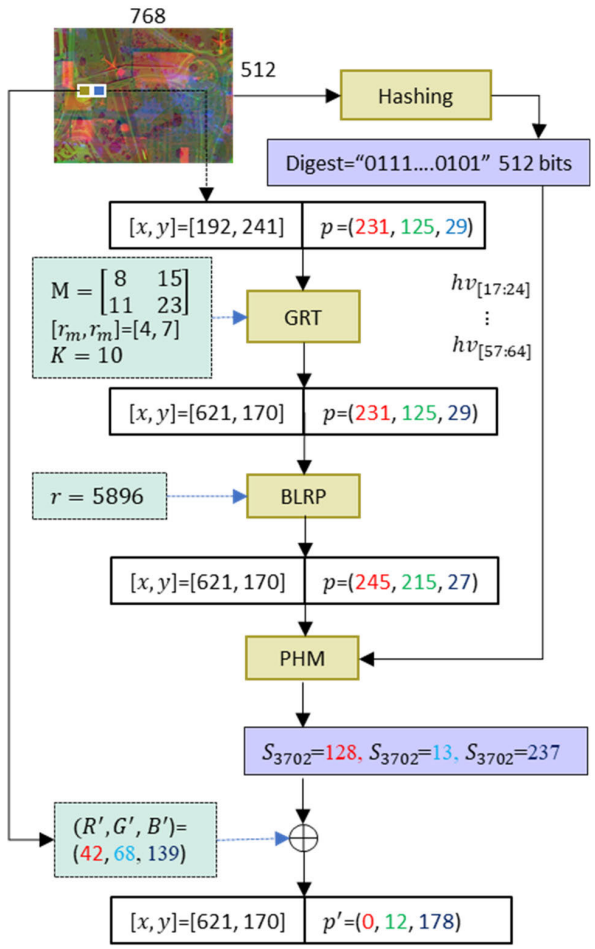


FIGURE 7. An example of image encryption produced by our GRTPHM scheme.

process to restore the chromatic channel values of the pixels. This is achieved using the pixel decryption expression define in (22). Since the secret keys are known, we can reproduce the same chaotic sequence, ensuring accurate decryption and restoration of the original pixel values.

$$p = \begin{cases} R_{i+1} = [(S_{3i} \oplus R'_{i+1}) - B'_i] \ mod \ 256 \\ G_{i+1} = [(S_{3i+1} \oplus G'_{i+1}) - R'_i] \ mod \ 256 \\ B_{i+1} = [(S_{3i+2} \oplus B'_{i+1}) - G'_i] \ mod \ 256 \end{cases} \quad (22)$$

Following our previous example, we utilize DEC PHM process to decrypt the pixel at the position $[x, y] = [621, 170]$. The pixel value $p' = (0, 12, 178)$ will be temporarily decrypted to the same position but with different value $p = (245, 215, 27)$. Then, we perform the inverse bit-level random permutation (IBLRP) to rearrange the bit positions with the aid of the secret key (key 5), thus re-shuffle bits to become $p = (231, 125, 29)$. Finally, we operate the inverse generalized rectangular transformation (IGRT) for K times by referring to the secret keys (keys 1-4). This produces the final deciphered pixel at the position $[x, y] = [192, 241]$ which has the pixel value $p = (231, 125, 29)$. The position and pixel value are identical to one stored in the superimposed pseudo image I_p.

VII. EXPERIMENTAL RESULTS AND ANALYSIS

We implemented our GRTPHM algorithm and collected our experiments using Python 3.8 on a personal computer equipped with a 2.80 GHz CPU, 16 GB RAM. When producing the PHM hyperchaotic sequence, we set the initial values $x_0 = -10, y_0 = 5, z_0 = 10, w_0 = 1, u_0 = -5$, and $k = 0.11$.

In this section, we present experimental results. First, we compare our algorithm exclusively with three rectangular transformation (RT) algorithms. Next, we evaluate and compare the image encryption performance using nine standard metrics: key space, visual perception, histogram, analysis, correlation, information entropy, local entropy, image sensitivity, key sensitivity, and speedup analysis.

Additionally, we provide experimental results for two types of null images, where all pixels are set to 0, and white images, where all pixels are set to 255. Null images are essential for evaluating whether an encryption algorithm can withstand one of the most malicious cryptographic attacks: the chosen-plaintext attack (CPA). Due to their uniform pixel distribution (all 0s or all 255s), null images provide an ideal test case for assessing an encryption scheme's ability to obscure predictable structures. If an encrypted null image exhibits discernible patterns or statistical weaknesses, it may indicate vulnerabilities exploitable by attackers. Therefore, analyzing the encryption results of null images helps determine whether the algorithm effectively diffuses information, eliminates plaintext predictability, and resists CPA-based cryptanalysis.

A. RECTANGULAR TRANSFORMATION COMPARISON

Table 10 presents a comparative analysis with three RT algorithms [12], [13], [14]. The rationale behind this comparison is that, to the best of our knowledge, there are only three existing image encryption algorithms in the literature that utilize rectangular transformation for encryption.

First, while GRTPHM scheme can correctly decrypt an image using inverse generalized rectangular transformation (IGRT), all other algorithms fail to do so because they rely on a naïve inverse matrix, which, as demonstrated in our example, is incapable of restoring the original pixel positions. For fairness in comparison, we evaluate only the encryption

TABLE 10. Comparative analysis with three RT algorithms.

Type	GRTPHM	[12]	[13]	[14]
Inverse RT	Success	Fail	Fail	Fail
Speed (s)	0.152	0.126	3.76	2.28
Key Space	$\approx 10^{250}$	$\approx 10^{77}$	$\approx 10^{102}$	$\approx 10^{117}$
Matrix Gen.	General	Special	Special	Special
Periodicity	Large	Moderate	Small	Small
Multiple Enc.	Support	X	X	X

speed using a test color image with a size of 6 MB. Our algorithm achieves real-time performance, with an encryption speed that is slightly slower than [12] but faster than the other two competing methods [13], [14].

Our scheme ensures security by requiring a large key space and leveraging generalized and versatile RT matrices, unlike competing algorithms that rely on specialized matrices with structural constraints (e.g., a fixed first element in a 2×2 matrix). Additionally, our approach maintains a high matrix periodicity, enhancing the effectiveness of inverse generalized rectangular transformation (IGRT) for accurate image decryption. A distinctive advantage of our scheme is its ability to faithfully reconstruct the original host image, a capability that most RT-based algorithms lack. Furthermore, our method supports multi-image encryption, enabling the construction of a large host image by superimposing multiple smaller images, whereas competing algorithms are restricted to single-image encryption. Overall, the comparative analysis confirms that our GRTPHM scheme surpasses the three competing methods.

B. KEY SPACE

Key space is a crucial metric to evaluate the security of an image encryption algorithm. If the key space exceeds 2^{100}, equivalent to 1.2677×10^{30}, the algorithm can resist brute force attacks. The comparison of key space with other 15 algorithms is shown in Table 11. The results show that our GRTPHM scheme has a sufficiently large key space to withstand the brute force attacks.

TABLE 11. Comparison of key space.

Algorithm	Key space	Algorithm	Key space
Our GRTPHM	2.8639×10^{250}	[23]	1.0000×10^{60}
[6]	7.9228×10^{124}	[24]	4.2950×10^{71}
[7]	$> 8.5782 \times 10^{256}$	[29]	3.7577×10^{109}
[4]	$> 8.0347 \times 10^{59}$	[32]	1.1579×10^{109}
[12]	1.158×10^{77}	[33]	$\cong \infty$
[13]	5.0000×10^{102}	[34]	3.9402×10^{115}
[14]	5.0000×10^{117}	[36]	1.8527×10^{78}
[22]	6.5536×10^{48}	[37]	7.2684×10^{134}

C. VISUAL PERCEPTION

Figure 8 shows the cipher images with different sizes encrypted by our GRTPHM scheme using various rectangular matrices. The details and contour features are hidden completely and no meaningful information can be unveiled from the noise-like cipher images. Note that our scheme can encrypt equilateral image as well as rectangular images, Fig. 8(a)-(d). In addition, our scheme can encrypt three grayscale images simultaneously by superimposing them into a pseudo color image, see Fig. 8(e), (f). Finally, we remark that GRTPHM is powerful to cipher all black and white image, Fig. 8(g), (h), indicating that it can resist the chosen plaintext attack (CPA) suggested in [25].

D. HISTOGRAM ANALYSIS

A good encryption algorithm should make the histogram of encrypted image to have an almost uniform distribution. Figure 9 exhibits the histogram of the original image and that of the ciphered image. The histogram of the encrypted image shows uniformly distributed among the interval [0, 255]. We use the chi-square test to quantitatively test the uniformity of histogram. The chi-square test has the form, as shown in (23):

$$\chi^2 = \sum_{i=0}^{255} \frac{(k_i - \overline{k})^2}{\overline{k}}, \tag{23}$$

where k_i represents the occurrence frequency of single channel i and $\overline{k}$ denotes the expected occurrence frequency of this channel level. Table 12 shows the chi-square statistics for different images including the all black and all white images. Assuming the significant level $\alpha = 0.05$, the corresponding critical value (cv) under the degree of freedom $df = 255$ is $\chi^2(0.05, 255) = 293.248$. Clearly, all chi-square statistics are less than the critical value, indicating that the histogram of the ciphertext image passes the testing of uniform distribution.

TABLE 12. Chi-square test, $df = 255$, $\alpha = 0.05$, $cv = 293.248$.

ID	Name	Red	Green	Blue	Result
1	Baboon	252.7520	266.0547	258.6172	Pass
2	Pepper	260.0156	253.8262	270.1211	Pass
3	Kod20	276.0308	212.8671	284.6754	Pass
4	Kod24	287.5066	271.5676	228.8913	Pass
5	Runway	272.4709	217.4028	248.2824	Pass
6	Bridge	230.8962	280.2801	258.2094	Pass
7	Black	290.2447	268.2482	276.9243	Pass
8	White	235.3723	255.7825	250.9456	Pass

E. CORRELATION ANALYSIS

A good encrypted algorithm should reduce the correlation between two adjacent pixels in horizontal, vertical, and diagonal directions. We introduce the correlation coefficient r_{xy} between two adjacent pixels, as shown in (24), where $\overline{x} = \frac{1}{N}\sum_{i=1}^{N} x_i$ and $\overline{y} = \frac{1}{N}\sum_{i=1}^{N} y_i$. Figure 10 shows the correlation diagrams for three directions. We can find that the correlation diagrams corresponding to the ciphertext images are random-like, indicating that our encryption algorithm has broken the neighboring pixel's information.

We chose $N = 8000$ pairs of adjacent pixels with stratification in three directions from the plaintext image and the cipher image. The derived r_{xy} is shown in Table 13. Observing these

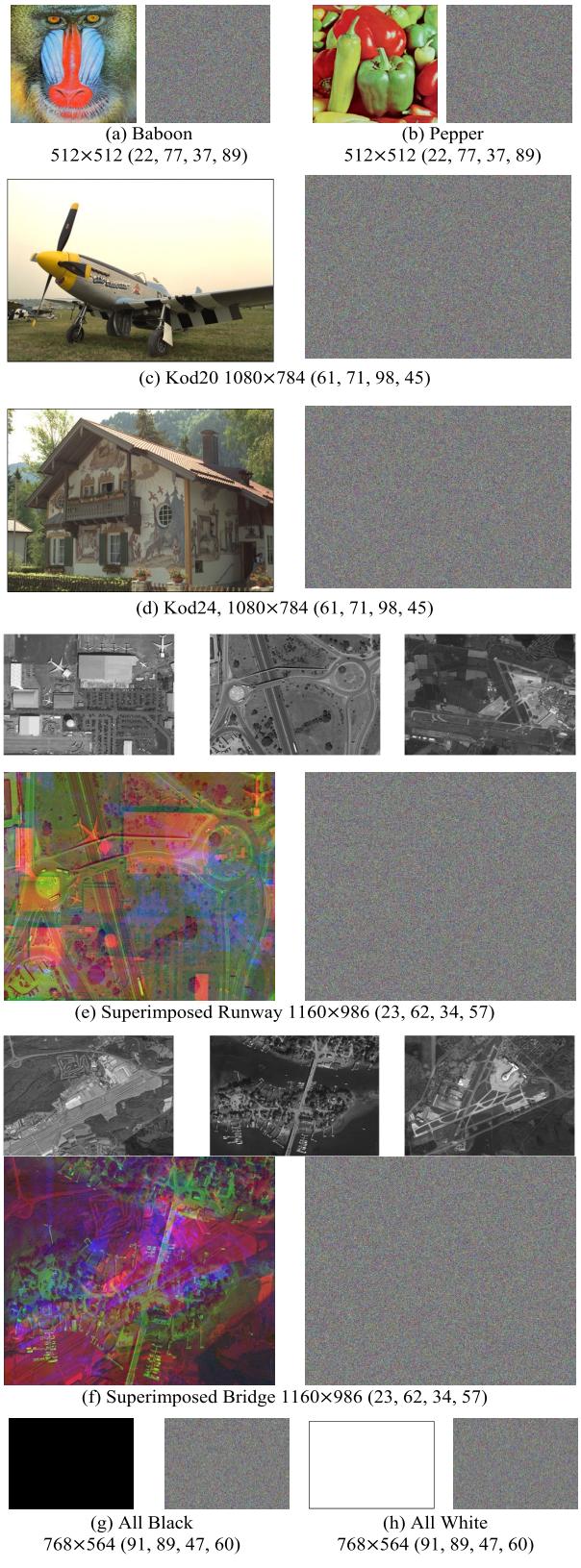


FIGURE 8. Visual perception for several plaintext (decrypted) and ciphered images.

statistics, we can find that r_{xy} in each direction of the cipher image are close to 0, which demonstrates that our algorithm

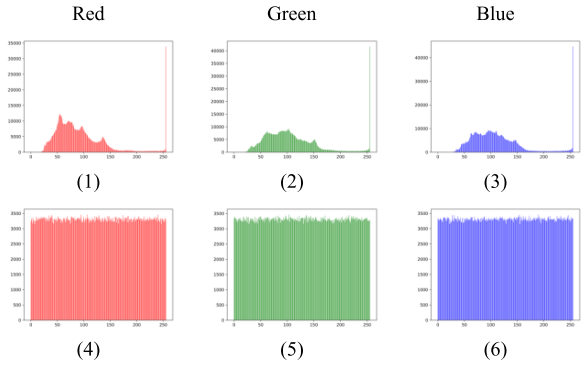


FIGURE 9. Histograms of the test image, Kod24, in three channels before the encryption (1)-(3) and those after the encryption (4)-(6).

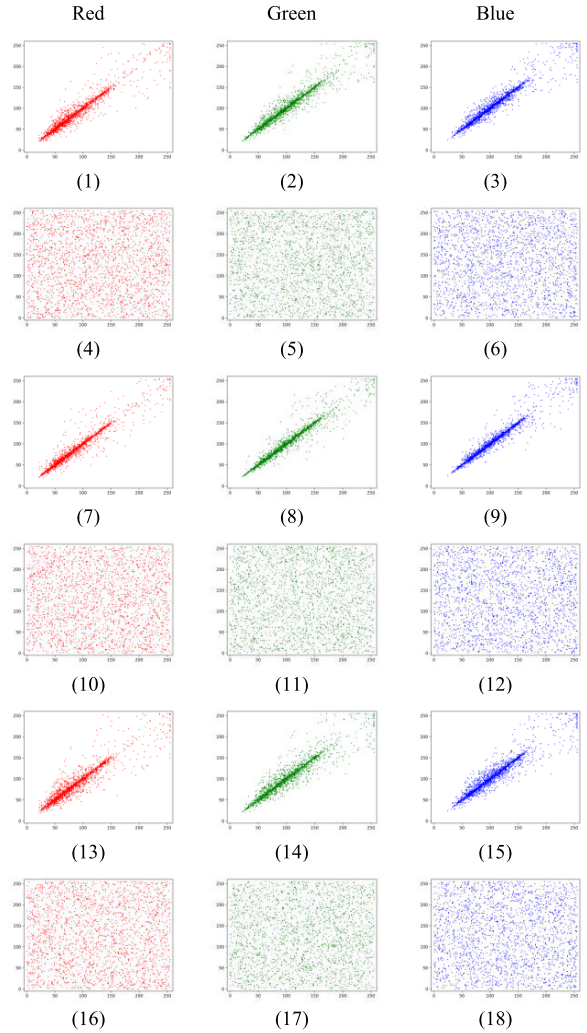


FIGURE 10. Adjacent pixels correlation diagrams in three chromatic channels before the encryption: (1)-(3) in the horizontal direction, (7)-(9) in the vertical direction, and (13)-(15) in the diagonal direction. Those after the encryption in three directions are shown in (4)-(6), (10)-(12), and (16)-(18), respectively.

can effectively resist statistical attacks.

$$r_{xy} = \frac{\sum_{i=1}^{N}(x_i - \overline{x})(y_i - \overline{y})}{\sqrt{\sum_{i=1}^{N}(x_i - \overline{x})^2 \cdot \sum_{i=1}^{N}(y_i - \overline{y})^2}}, \tag{24}$$

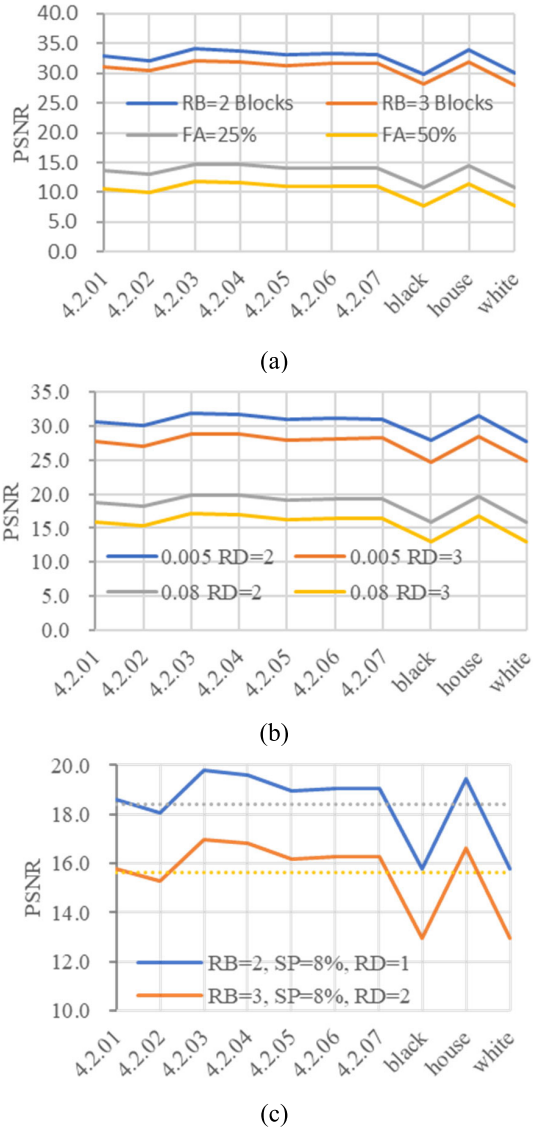


FIGURE 11. PSNR values for the robustness test. (a) Type I and II attacks; (b) Type III attack; (c) Type IV attack.

F. INFORMATION ENTROPY ANALYSIS

Information entropy can measure the degree of uncertainty and randomness of information source. For a color image, $H(m)$ is used to measure the confusion of individual channel and the pixel value distribution at this chromatic channel, as shown in (25), where $p(m_i)$ represents the probability of m_i and $L = 256$. The theoretical value at each channel is 8 when the occurrence frequency of the pixel value at each channel is equal.

$$H(m) = -\sum_{i=0}^{L-1} p(m_i) \log_2 p(m_i) \tag{25}$$

Table 14 lists the information entropy of each channel of the encrypted images. Observing the statistics, we can find that the information entropy at each channel is close to 8, demonstrating that our encrypted algorithm has very good performance. In addition, Table 15 compares the average information entropy on three channels with other encryption schemes. The statistics show that our scheme has larger information entropy than those of other counterparts'. Those mean that our GRTPHM has a good property of information entropy.

TABLE 13. A comparison of correlation (r_{xy}) of two adjacent pixels.

	Orientation	R channel	G channel	B channel
Plaintext	Horizontal	0.933660	0.950104	0.898976
	Vertical	0.974600	0.969372	0.936118
	Diagonal	0.929915	0.928802	0.848570
Our GRTPHM	Horizontal	0.032000	-0.017000	0.015000
	Vertical	-0.007600	-0.038800	0.011000
	Diagonal	-0.020100	-0.025300	0.030700
[17]	Horizontal	0.933660	0.950104	0.898976
	Vertical	0.974600	0.969372	0.936118
	Diagonal	0.929915	0.928802	0.848570
[18]	Horizontal	0.933660	0.950104	0.898976
	Vertical	0.974600	0.969372	0.936118
	Diagonal	0.929915	0.928802	0.848570
[19]	Horizontal	0.006454	0.001801	0.009926
	Vertical	0.005386	0.004798	0.004394
	Diagonal	0.006142	0.002753	0.003562
[20]	Horizontal	0.004900	0.005400	0.005300
	Vertical	0.003100	0.000100	0.002200
	Diagonal	0.000700	0.001700	0.000700
[13]	Horizontal	0.000538	0.001186	-0.002372
	Vertical	-0.007058	0.000177	0.007818
	Diagonal	0.000573	-0.001693	-0.000927
[14]	Horizontal	0.000538	0.001186	-0.002372
	Vertical	-0.007058	0.000177	0.007818
	Diagonal	0.000573	-0.001693	-0.000927
[29]	Horizontal	0.001600	-0.001600	-0.000200
	Vertical	0.001900	0.002700	-0.002500
	Diagonal	0.001500	0.002600	-0.001500
[32]	Horizontal	0.003700	0.000400	-0.004900
	Vertical	0.003000	0.001900	-0.102000
	Diagonal	0.003000	0.004500	-0.003300
[33]	Horizontal	-0.003110	-0.001030	0.002010
	Vertical	-0.006420	0.002460	0.001810
	Diagonal	0.000550	-0.006730	0.000390
[34]	Horizontal	-0.003260	0.000140	0.001820
	Vertical	-0.002880	0.000230	0.000380
	Diagonal	0.000430	-0.001420	0.001380
[36]	Horizontal	-0.000440	-0.009700	0.001800
	Vertical	-0.000570	-0.004400	0.004200
	Diagonal	-0.000060	0.000500	0.006200
[37]	Horizontal	-0.030900	0.003600	-0.024800
	Vertical	0.018600	0.047100	0.001600
	Diagonal	0.016500	0.010500	0.005700

TABLE 14. Information entropy test.

ID	Name	Red	Green	Blue
1	Baboon	7.999303	7.999268	7.999286
2	Pepper	7.999283	7.999302	7.999254
3	Kod20	7.999765	7.999819	7.999757
4	Kod24	7.999755	7.999768	7.999805
5	Runway	7.999828	7.999863	7.999843
6	Bridge	7.999854	7.999823	7.999837
7	Black	7.999516	7.999553	7.999538
8	White	7.999609	7.999573	7.999582

G. LOCAL ENTROPY ANALYSIS

We introduce the local Shannon entropy to measure the randomness of the ciphertext image. As suggested in [26], the local Shannon entropy can be expressed in (26), where $\{s_i | i = 1, 2, \ldots, \}K$ represent a non-overlapping blocks selected

TABLE 15. Information entropy comparison.

Algorithms	Baboon	Girl	Couple
Our GRTPHM	7.999739	7.999138	7.999150
[17]	7.991296	7.991575	991349
[20]	7.997200	N/A	N/A
[13]	7.998973	7.998989	7.999136
[14]	7.999784	7.998991	7.999001
[29]	7.999300	7.998300	7.997300
[32]	7.994900	7.997400	7.991800
[33]	7.999390	7.999320	7.999310
[34]	7.999528	7.999557	7.999497
[36]	7.999500	7.999200	7.999300
[37]	7.999300	7.999200	7.999600

with T_B pixels from the test image, and $H(s_i)$ represents the corresponding information entropy at i-th block derived from (25).

$$\overline{H_{(k,T_B)}}(s) = \sum_{i=1}^{K} \frac{H(s_i)}{K} \tag{26}$$

In our test, we select $K = 30$ blocks with stratification and $T_B = 1936$ pixels, complying with the settings suggested in [26]. Table 16 shows the confidence interval of such setting and the local Shannon entropy collected using our encryption algorithm. It can see that our GRTPHM algorithm passes the tests, offering a random-like ciphertext image.

TABLE 16. Local Shannon entropy test, $\alpha = 0.05$ Confidence Interval [7.901901305, 7.903037329].

ID	Name	Red	Green	Blue	Status
1	Baboon	7.902467	7.901979	7.902482	Pass
2	Pepper	7.902144	7.902014	7.901971	Pass
3	Kod20	7.902962	7.902437	7.902576	Pass
4	Kod24	7.901910	7.902541	7.901983	Pass
5	Runway	7.902430	7.902360	7.901958	Pass
6	Bridge	7.902530	7.903007	7.902560	Pass
7	Black	7.902908	7.902384	7.901982	Pass
8	White	7.901927	7.901966	7.902810	Pass

H. IMAGE SENSITIVITY

The aim of encryption is to obtain random-like ciphertext images such that attackers cannot figure out the internal relations between plaintext and ciphertext. An attacker may conduct the chosen-plaintext attack, also known as differential attack. An attacker may first possess a ciphertext image, C^1, encrypted from the plaintext image, P^1. Next, an attacker may produce the second one, C^2, which is derived by encrypting P^2, where P^1 and P^2 have one-bit difference. By comparing C^1 and C^2, an attacker may observe and even derive a meaningful relationship between them, thus breaking the cryptographic system. Thus, a good encryption algorithm should offer high image sensitivity able to resist the chosen-plaintext attack.

Number of Pixels Change Rate (NPCR) and Unified Average Changing Intensity (UACI) are two metrics used to test the sensitivity of an encryption algorithm to plaintext image [27], [28]. They can be derived by (27)-(29), where $C^1_{R,G,B}(i,j)$ and $C^2_{R,G,B}(i,j)$ are two ciphertext images and F represents the largest allowed pixel value in a channel.

$$NPCR = \frac{1}{M \times N}[\sum_{i,j} D_{R,G,B}(i,j)] \times 100\%, \tag{27}$$

$$UACI = \frac{1}{M \times N}\left[\sum_{i,j} \frac{\left|C^1_{R,G,B}(i,j) - C^2_{R,G,B}(i,j)\right|}{F}\right] \times 100\%, \tag{28}$$

$$D_{R,G,B}(i,j) = \begin{cases} 0 & if\ C^1_{R,G,B}(i,j) = C^2_{R,G,B}(i,j) \\ 1 & if\ C^1_{R,G,B}(i,j) \neq C^2_{R,G,B}(i,j), \end{cases} \tag{29}$$

Statistical hypothesis NPCR and UACI tests were introduced to derive the critical value N^*_α for NPCR, as shown in (30) and critical range $(U^{*-}_\alpha, U^{*+}_\alpha)$ for UACI, as shown in (31)-(32).

$$N^*_\alpha = \frac{F - \Phi^{-1}(\alpha)\sqrt{\frac{F}{MN}}}{F+1}. \tag{30}$$

$$U^{*-}_\alpha = \frac{F+2}{3(F+1)} - \frac{1}{\sqrt{MN}}\Phi^{-1}(\frac{\alpha}{2})\sqrt{\frac{(F+2)(F^2+2F+3)}{18F(F+1)^2}}. \tag{31}$$

$$U^{*+}_\alpha = \frac{F+2}{3(F+1)} + \frac{1}{\sqrt{MN}}\Phi^{-1}(\frac{\alpha}{2})\sqrt{\frac{(F+2)(F^2+2F+3)}{18F(F+1)^2}}. \tag{32}$$

In these equations, F represents the largest allowed integer related to image format; for example, $F = 255$ for a grayscale image or a channel of the color image, and $\Phi^{-1}(\cdot)$ denotes the inverse cumulative density function (CDF) of the standard Normal distribution $\mathbb{N}(0, 1)$.

Given two ciphertext images, C^1, C^2, if NPCR $\geq N^*_\alpha$, we accept the null hypothesis (H_0), indicating that C^2 is a random-like encrypted image. Similar, if UACI $\in (U^{*-}_\alpha, U^{*+}_\alpha)$, we accept the null hypothesis, indicating that C^2 is a random-like encrypted image. Furthermore, if both criteria about NPCR and UACI are satisfied, we can conclude that C^2 is able to resist the chosen-plain attack.

Table 17 lists the N^*_α and $(U^{*-}_\alpha, U^{*+}_\alpha)$ corresponding to our test images under the condition significant level, $\alpha = 0.05$.Note that they depend not only on α but also on the image resolutions.

Table 18 shows the NPCR test results and Table 19 lists the UACI test results under the same significant level. Observing the statistics, we find that all values satisfy the conditions, thus passing the tests. We conclude that our GRTPHM algorithm can resist the chosen-plaintext attack.

Table 20 presents a comparison of NPCR results across different encryption algorithms, while Table 21 provides the corresponding UACI comparison. For consistency, we converted all test images to grayscale and change one bit of

TABLE 17. Critical values and critical ranges for NPCR and UACI tests $\alpha = 0.05, F = 255$.

ID	Name	M	N	N_α^*	$(U_\alpha^{*-}, U_\alpha^{*+})$
1	Baboon	512	512	0.995893	[0.333730, 0.335541]
2	Pepper	512	512	0.995893	[0.333730, 0.335541]
3	Kod20	1080	784	0.995982	[0.334131, 0.335139]
4	Kod24	1080	784	0.995982	[0.334131, 0.335139]
5	Runway	1160	986	0.995998	[0.334202, 0.335069]
6	Bridge	1160	986	0.995998	[0.334202, 0.335069]
7	Black	768	564	0.995938	[0.333931, 0.335340]
8	White	768	564	0.995938	[0.333931, 0.335340]

TABLE 18. NPCR results in the image sensitive test ($\alpha = 0.05$).

ID	Name	Red	Green	Blue	Test
1	Baboon	0.995953	0.996124	0.996269	Pass
2	Pepper	0.996159	0.996300	0.995953	Pass
3	Kod20	0.995989	0.995983	0.996090	Pass
4	Kod24	0.996021	0.996179	0.996120	Pass
5	Runway	0.996131	0.996073	0.996173	Pass
6	Bridge	0.996118	0.996028	0.996100	Pass
7	Black	0.996094	0.996126	0.996198	Pass
8	White	0.996172	0.996352	0.996135	Pass

TABLE 19. UACI results in the image sensitive test ($\alpha = 0.05$).

ID	Name	Red	Green	Blue	Test
1	Baboon	0.334694	0.335439	0.334802	Pass
2	Pepper	0.334885	0.334648	0.334049	Pass
3	Kod20	0.334782	0.334907	0.334496	Pass
4	Kod24	0.334974	0.334807	0.335126	Pass
5	Runway	0.334654	0.334491	0.335037	Pass
6	Bridge	0.334952	0.334591	0.334613	Pass
7	Black	0.334792	0.334809	0.334827	Pass
8	White	0.334490	0.334838	0.334561	Pass

pixel value randomly before collecting the results. Statistical analysis shows that most NPCR and UACI values pass the $\alpha = 0.05$ level of the hypothesis test.

It is important to note that one cannot conclusively determine that an encryption algorithm A is superior to another algorithm B solely based on the magnitude of their NPCR and UACI scores if both pass the hypothesis test. In such cases, it can be inferred that both algorithms A and B can generate random-like ciphertext images, with any differences in test scores arising purely from stochastic processes [28].

I. KEY SENSITIVITY

An encryption algorithm must exhibit key sensitivity, meaning that two keys differing by only a single bit should produce ciphertext images with significant differences. To evaluate the key sensitivity of our GRTPHM algorithm, we randomly selected two initial values x_0 and x_0', differing by one bit, and analyzed the results using NPCR and UACI metrics.

Table 22 and Table 23 show the evaluation results. The data indicate that all statistics meet the criteria of NPCR and UACI hypothesis testing. This demonstrates that even minor variations in secret keys result in substantial differences

TABLE 20. NPCR comparison for different encryption schemes.

NPCR	Baboon	Peppers	Kod20	Black	White
Our	0.996060	0.996076	0.996011	0.996089	0.996149
[12]	0.996208	-	-	0.996465	0.996207
[3]	0.996070	-	-	-	-
[4]	0.996389	-	-	0.996171	0.996294
[14]	1.000000	1.000000	-	-	-
[18]	0.994300	0.996400	-	-	-
[20]	0.996500	0.996300	-	-	-
[13]	0.997200	0.997400	-	-	-
[21]	0.996000	0.995900	-	-	-
[29]	0.995972	0.996117	-	-	-
[32]	0.996092	0.996048	0.996109	-	-
[33]	0.995968	0.996052	0.996029	-	-
[34]	0.995878	0.995812	0.996100	-	-
[36]	0.996095	-	-	-	-
[37]	0.996223	0.996012	0.996123	-	-

TABLE 21. UACI comparison for different encryption schemes.

UACI	Baboon 4.2.03	Peppers 4.2.07	Kod20	Black	White
Our	0.334978	0.334527	0.334728	0.334809	0.334630
[12]	0.335034	-	-	0.334641	0.334721
[3]	0.335236	-	-	-	-
[4]	0.334515	-	-	0.334574	0.334582
[14]	0.334600	0.334200	-	-	-
[18]	0.335300	0.335300	-	-	-
[20]	0.335000	0.334700	-	-	-
[13]	0.334900	0.335300	-	-	-
[21]	0.335700	0.335100	-	-	-
[29]	0.334380	0.334614	-	-	-
[32]	0.334712	0.334779	0.334056	-	-
[33]	0.334637	0.334255	0.334695	-	-
[34]	0.331326	0.335318	0.334352	-	-
[36]	0.334671	-	-	-	-
[37]	0.334513	0.335020	0.333812	-	-

in the corresponding ciphertext images. Consequently, our proposed scheme has been proven to be highly sensitive to changes in the secret key.

TABLE 22. NPCR results in the key sensitive test.

ID	Name	Red	Green	Blue	Status
1	Baboon	0.995941	0.996037	0.995937	Pass
2	Pepper	0.996288	0.996021	0.996281	Pass
3	Kod20	0.995947	0.996026	0.996146	Pass
4	Kod24	0.996111	0.996166	0.995982	Pass
5	Runway	0.996163	0.996148	0.996057	Pass
6	Bridge	0.995985	0.996080	0.996079	Pass
7	Black	0.996020	0.996265	0.996255	Pass
8	White	0.996237	0.996221	0.996161	Pass

TABLE 23. UACI results in the key sensitive test.

ID	Name	Red	Green	Blue	Status
1	Baboon	0.334839	0.334547	0.334674	Pass
2	Pepper	0.334379	0.334269	0.335011	Pass
3	Kod20	0.334464	0.334498	0.334862	Pass
4	Kod24	0.334908	0.334172	0.334789	Pass
5	Runway	0.334102	0.334970	0.334833	Pass
6	Bridge	0.334558	0.334597	0.334969	Pass
7	Black	0.334048	0.334913	0.334883	Pass
8	White	0.334762	0.334704	0.334643	Pass

J. ROBUSTNESS ANALYSIS

Image corruption can occur due to noisy transmission environments or malicious attacks. Therefore, an image encryption algorithm must account for robustness to ensure accurate reconstruction of the encrypted image. To evaluate the robustness of our algorithm, we employ Peak Signal-to-Noise Ratio (PSNR) and the Structural Similarity Index Measure (SSIM) as quantitative metrics.

Let I_O represent the original image and I_E the ciphered image produced by our scheme. To evaluate the robustness of our encryption method, we subject I_E to four types of malicious attacks as part of our robustness test.

Type I: Random Area Cropping Attack

In this attack, we randomly replace two blocks (RB=2) or three blocks (RB=3) in I_E. Each block is a 20×20 pixel square filled with black pixels, simulating partial data loss at unpredictable locations.

Type II: Fixed Position Cropping Attack

This attack involves cropping a fixed area from I_E, proportional to the total image size. The cropped regions are predetermined rather than randomly selected. We evaluate two levels of cropping: FA=25% and FA=50% of the encrypted image, assessing the algorithm's ability to reconstruct the image under different degrees of data loss.

Type III: Salt & Pepper Attack

In this attack, we introduce salt & pepper noise to I_E, where a certain percentage of pixels are randomly altered to either the minimum (black) or maximum (white) intensity values. We consider two noise levels: SP=0.5% and SP=8% of the total pixels in the image. Additionally, we perform multiple rounds of noise application to simulate cumulative degradation. Specifically, we conduct either two rounds (RD=2) or three rounds (RD=3) of noise addition. This results in an approximate total noise level of 1.0% for RD=2 and 24% for RD=3, respectively. This approach allows us to assess the resilience of our encryption scheme under progressively severe noise disturbances.

Type IV: Hybrid Attack

This attack combines Type I and Type III, making it a more severe test of our encryption scheme's robustness. The attack process follows a sequential approach. First, we conduct the Type I attack with two blocks (RB=2) or three blocks (RB=3). Next, we introduce the Type III attack with SP=8%. Finally, we impose RD=1 round for RB=2 blocks and RD=2 rounds for RB=3 blocks, leading to a progressively severe distortion effect. By combining cropping-induced data loss with intensive noise contamination, this hybrid attack rigorously tests the resilience of our encryption scheme in extreme conditions.

In each test, let I'_E represent the ciphered image and I'_A represent the same image after undergoing one f the four attacks. Additionally, let I'_{DA} denote the deciphered image obtained from I'_A, while I_O represents the deciphered image from I'_E, which is identical to the original plaintext image. To evaluate the robustness of our encryption scheme, we compute the Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) between I_O and I'_{DA}. A higher PSNR value and an SSIM score closer to 1.0 indicate better robustness in recovering the original image despite attacks. The test dataset comprises eight 512×512 standard test images from the SIPI image database, along with two null images—a completely black image and a completely white image. These diverse test cases ensure a comprehensive evaluation of the algorithm's performance across different image characteristics, including variations in texture, contrast, and uniformity.

Figure 11 shows the PSNR values for four types of attack. As shown in Fig. 11(a), the average PSNR value in Type I attack exceeds 30 dB, indicating that our algorithm maintains relatively high image quality under this attack. The PSNR value of the type II attack drops significantly to around 10–13 dB, demonstrating that larger cropped areas have a stronger impact on reconstruction accuracy.

With regard to the Type III attack, as depicted in Fig. 11(b), higher noise ratios degrade image quality, leading to lower PSNR values. This confirms that increasing the percentage of salt & pepper noise directly impacts the deciphered image's fidelity. Finally, Fig. 11(c) shows that the PSNR value is highly influenced by the number of noise rounds. When RD=2, the average PSNR is 18.41 dB, whereas for RD=3, it further declines to 15.61 dB, indicating that multiple rounds of noise severely affect decryption accuracy.

These results demonstrate the varying degrees of robustness of our encryption scheme under different attack scenarios.

Table 24 to 25 present the SSIM values for four the four types of attacks. As observed in the results, Type I attack has only a minor impact on SSIM values, particularly when RB=2 blocks, indicating that our scheme can effectively reconstruct the image despite small, randomly removed regions. Type II attack has a severe effect on SSIM values, especially when FA=50%, due to the substantial loss of contiguous pixel information, making accurate reconstruction more challenging.

TABLE 24. SSIM results for the robustness test: type I and II attacks.

Test Image	Type I RB=2	Type I RB=3	Type II FA=25%	Type II FA=50%
4.2.01	0.935596	0.902156	0.079583	0.039627
4.2.02	0.936980	0.910190	0.085422	0.039747
4.2.03	0.976184	0.965009	0.330661	0.164635
4.2.04	0.949401	0.923571	0.138088	0.064763
4.2.05	0.944670	0.925454	0.140131	0.070935
4.2.06	0.957781	0.944749	0.197313	0.098890
4.2.07	0.947757	0.929283	0.133626	0.063221
black	0.893031	0.860370	0.000119	0.000015
house	0.960249	0.934595	0.201059	0.098631
white	0.902209	0.855623	0.013828	0.008347
Ave.	0.940386	0.915100	0.131983	0.064881

Despite the introduction of noise in Type III attack, our scheme maintains moderate SSIM values, suggesting that it can withstand certain levels of noise interference while still preserving image structure. The robustness tests reveal

that cropping attacks have a more severe impact than salt & pepper noise attacks. This is because cropping results in the permanent loss of large contiguous pixel regions, whereas salt & pepper noise affects individual pixels in a scattered manner, allowing for partial recovery of image details.

These findings confirm the resilience of our encryption scheme under various attack scenarios, with cropping posing a more significant challenge due to irreversible data loss.

TABLE 25. SSIM results for the robustness test: type III attack.

Test Image	RD=2 SP=0.5%	RD=3 SP=0.5%	RD=2 SP=8%	RD=3 SP=8%
4.2.01	0.802751	0.802751	0.2566321	0.134236
4.2.02	0.810673	0.810673	0.2779568	0.151107
4.2.03	0.924115	0.924115	0.6186874	0.467558
4.2.04	0.849308	0.849308	0.3731817	0.225575
4.2.05	0.832598	0.832598	0.3453071	0.214452
4.2.06	0.867068	0.867068	0.4442063	0.295034
4.2.07	0.846531	0.846531	0.3626771	0.216882
black	0.685810	0.685810	0.0525708	0.003649
house	0.868012	0.868012	0.4460349	0.301577
white	0.709846	0.709846	0.0949240	0.028148
Ave.	0.819671	0.819671	0.3272178	0.203822

TABLE 26. SSIM results for the robustness test: type IV attack.

Test Image	Hybrid RB=2, SP=8%, RD=2		Hybrid RB=3, SP=8%, RD=3	
	PSNR	SSIM	PSNR	
4.2.01	18.59	0.249257	15.80	0.131920
4.2.02	18.06	0.272275	15.27	0.146682
4.2.03	19.78	0.613727	16.98	0.460161
4.2.04	19.60	0.363936	16.81	0.221237
4.2.05	18.96	0.338204	16.19	0.211167
4.2.06	19.03	0.431017	16.25	0.291332
4.2.07	19.07	0.352851	16.26	0.212606
black	15.76	0.048413	12.93	0.003050
house	19.47	0.437789	16.64	0.296087
white	15.75	0.088468	12.96	0.027230
Ave.	18.41	0.319594	15.61	0.200147

Figure 12 shows the deciphered images obtained after the encrypted images were subjected to the four types of malicious attacks for robustness testing. The results demonstrate that our GRTPHM scheme is capable of maintaining satisfactory visual perception even under various severe attack scenarios. Despite the impact of cropping and noise disturbances, the decrypted images retain recognizable structural details, highlighting the resilience of our encryption algorithm in preserving image integrity after decryption.

Figure 13 illustrates the PSNR values of the deciphered image when the encrypted image is subjected to a fixed position cropping attack, where the cropped area is measured as a percentage of the total image resolution. The results indicate a clear inverse relationship between cropping ratio and PSNR values—as the cropped area increases, the PSNR values decrease. Specifically, when 75% of the encrypted image is cropped, the decrypted image achieves a PSNR value of 8.09 dB, highlighting the substantial impact of large-area cropping on image reconstruction quality.

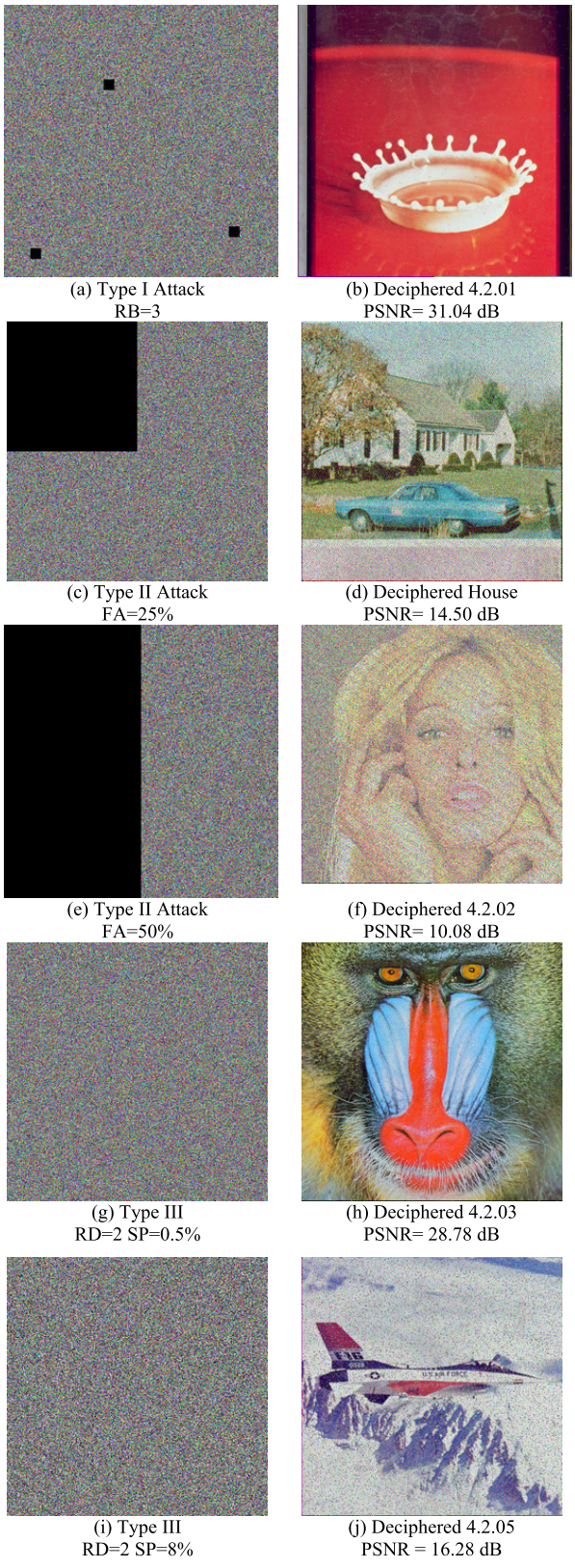
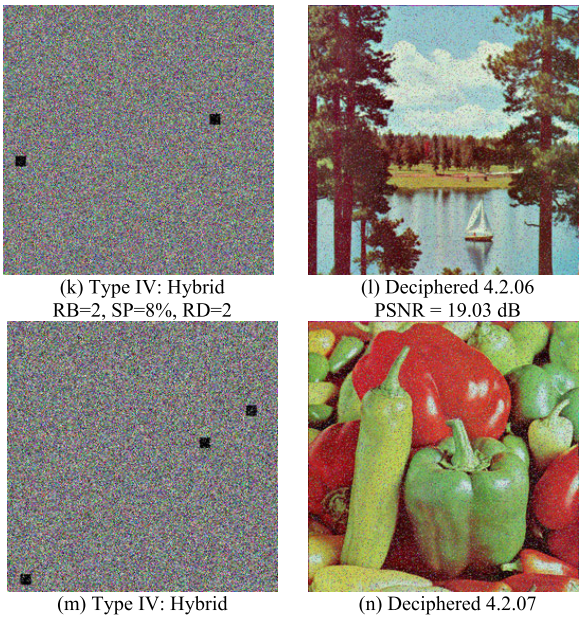


FIGURE 12. Deciphered images under malicious four types of attacks.



(k) Type IV: Hybrid RB=2, SP=8%, RD=2

(l) Deciphered 4.2.06 PSNR = 19.03 dB

(m) Type IV: Hybrid RB=3, SP=8%, RD=3

(n) Deciphered 4.2.07 PSNR = 16.26 dB

FIGURE 12. *(Continued.)* **Deciphered images under malicious four types of attacks.**

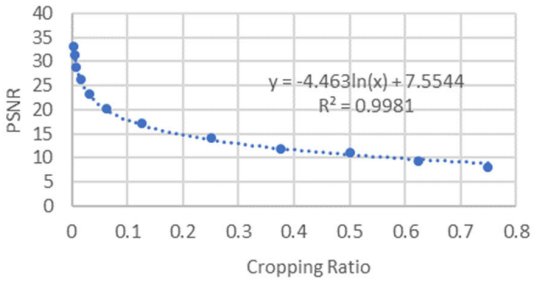


FIGURE 13. **The PSNR value corresponding to the fixed position cropping attack.**

K. SPEED ANALYSIS

Efficient image encryption is essential for supporting a wide range of real-time applications, with equally critical emphasis on the efficiency of image decryption. Table 27 details the runtime of our GRTPHM algorithm, measured in seconds over 120 executions, encompassing both the image permutation and diffusion phases.

TABLE 27. **A Comparison of the average timing for Image encryption and decryption.**

	$M \times N$	Matrix	GRT	FGRT	IGRT	Speedup
1	1080×784	$\begin{bmatrix} 61 & 71 \\ 98 & 45 \end{bmatrix}$	0.397	342.164	1.425	240.12
2	1160×986	$\begin{bmatrix} 23 & 62 \\ 34 & 57 \end{bmatrix}$	0.482	340.171	1.596	213.14
3	768×564	$\begin{bmatrix} 91 & 89 \\ 47 & 60 \end{bmatrix}$	0.251	270.569	1.304	207.49
4	1204×496	$\begin{bmatrix} 37 & 17 \\ 124 & 57 \end{bmatrix}$	0.344	78.707	1.330	59.18
5	1228×256	$\begin{bmatrix} 37 & 17 \\ 128 & 61 \end{bmatrix}$	0.227	55.449	1.319	42.04

Compared to decryption via forward encryption (FGRT), our algorithm achieves a significant acceleration by utilizing the inverse generalized rectangular transformation (IGRT). This optimization leads to speedup factors ranging from 42.04 to 240.12, demonstrating the substantial efficiency gain in the decryption process.

Figure 14 illustrates the speedup achieved by our inverse generalized rectangular transformation (IGRT) in relation to the periodicity of the rectangular matrix. The results indicate a clear trend: as the matrix periodicity increases, the speedup factor of IGRT also improves. Specifically, when the periodicity reaches 2600, our IGRT achieves a decryption speedup of 299.56 times, highlighting the efficiency advantage of leveraging high-periodicity matrices for accelerated decryption.

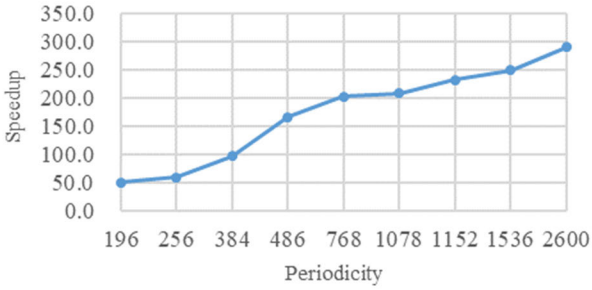


FIGURE 14. **The speedup benefits of the inverse GRT (IGRT) over forward GRT (FGRT) during image decryption process.**

The analysis reveals that the time complexity for decryption in our scheme is $O(M \times N \times K)$, outperforming the $O(M \times N \times (P - K))$ complexity of FGRT, where the periodicity (P) far exceeds the encryption iteration count (K). Although the decryption process (IGRT) generally requires more time than encryption (GRT) due to its four computational stages, our scheme ensures a substantial advantage by enabling fast and efficient image encryption and decryption, making it well-suited for practical applications requiring high-speed security processing.

VIII. CONCLUSION AND FUTURE WORKS

This study introduces GRTPHM, a novel generalized rectangular multi-image encryption scheme designed to address the challenges of modern image encryption. By leveraging the Generalized Rectangular Transform (GRT) for pixel permutation, GRTPHM effectively disrupts spatial correlations between neighboring pixels. The adaptive transformation matrix ensures flexibility, while the inverse generalized rectangular transform (IGRT) eliminates the need for prior knowledge of the transformation matrix's periodicity, overcoming the limitations of existing decryption methods. This innovation achieves an unprecedented speedup of up to 299 times compared to state-of-the-art techniques.

The Penta-Hyperchaotic Map (PHM), enhanced by a refined 5D discrete memristor- and meminductor-based hyperchaotic map, drives the pixel diffusion phase by generating high-entropy sequences. This ensures uniform randomness in encrypted images, significantly strengthening

resistance against statistical and differential attacks. Moreover, PHM enhances encryption key sensitivity, minimizing susceptibility to brute-force and cryptographic attacks. Additionally, the Bit-Level Random Permutation (BLRP) mechanism further improves diffusion at a granular level, ensuring that minor changes in the plaintext produce substantial variations in the ciphertext.

Extensive robustness testing was conducted to rigorously assess GRTPHM's resilience against various malicious attacks, including random and fixed-position cropping, salt & pepper noise, and hybrid distortions. The results confirm that GRTPHM maintains satisfactory decryption accuracy and visual integrity, even under extreme image corruption scenarios. Notably, our scheme demonstrates greater robustness against salt & pepper noise than cropping-based attacks, as the latter causes permanent information loss. The PSNR and SSIM analyses further validate that GRTPHM effectively mitigates degradation effects and preserves image quality under severe attack conditions. These comprehensive evaluations highlight the strong adaptability and fault tolerance of our encryption scheme, reinforcing its practicality for real-world security applications.

Additionally, testing using standard evaluation metrics, including null images composed entirely of black or white pixels, demonstrates GRTPHM's scalability and adaptability for encrypting both equilateral and rectangular images. Unlike conventional schemes, GRTPHM excels in both encryption strength and precise decryption across diverse image dimensions. Moreover, the multi-image encryption strategy allows multiple images to be securely encoded within a single transformation process, enhancing both security and efficiency by supporting composite encryption structures.

Comparative analyses confirm GRTPHM's superiority in accuracy, computational efficiency, and security, particularly in defending against sophisticated chosen-plaintext attacks. Through rigorous evaluation and experimental validation, GRTPHM emerges as a versatile and reliable encryption solution, offering enhanced performance, superior diffusion, and resilience against contemporary cyber threats.

While GRTPHM has demonstrated strong resilience against cropping, noise, and hybrid attacks, future work could explore its resistance to more sophisticated adversarial attacks, such as deep-learning-based cryptanalysis or adaptive chosen-ciphertext attacks. Investigating novel error-recovery mechanisms or incorporating redundant encoding techniques could further enhance the algorithm's robustness in highly adversarial environments. The current implementation of the Penta-Hyperchaotic Map (PHM) relies on the Euler method. Future work could explore more advanced numerical integration techniques to achieve higher precision and better preservation of chaotic properties.

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